

Kernel-Preconditioned Burer–Monteiro Flow: A Spectral Implicit Bias for Wide Networks Trained with Triplet Loss — Maxinfo Companion —

Comprehensive Data Companion: All Experimental Results

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Front note. Non-anonymous data-dump companion to the main paper. All numerical results from the underlying experiments are surfaced here for reference. The main paper preserves the theoretical contributions; this companion makes the experimental evidence fully transparent. The theoretical content (Sections 1–5) is identical in claim and proof structure to the main submission; the contribution of this document is the comprehensive experimental dump gathered into Appendices B–K. Source repository: <https://github.com/alisaffarini/spectral-bias>.

Abstract

We show that gradient flow on a wide ReLU embedding network, trained with a twice-differentiable loss on its Gram matrix $G = FF^\top$, evolves under a *kernel-preconditioned Burer–Monteiro flow* $\dot{G} = -2(KEG + GEK)$, where $E = \partial L / \partial G$ and K is the depth- L neural tangent kernel at the inputs. When $K = I$ this recovers the classical Burer–Monteiro dynamics for low-rank semidefinite optimization. Beyond this equivalence, the kernel form yields a *spectral implicit bias*: in the eigenbasis of K , mode (i, j) of G evolves at instantaneous rate $\lambda_i + \lambda_j$, and at NTK initialization the per-mode displacement $|\tilde{G}_{ii}(T) - \tilde{G}_{ii}(0)|$ scales as λ_i^2 . This is a sharp, falsifiable scaling law that does not follow from the NTK ODE without the BM viewpoint. We verify the slope-2 prediction across 30 multi-seed configurations: synthetic spheres ($N \in \{30, 100, 1000\}$, depth $L \in \{1, 2\}$) yielding fitted slopes 1.38 to 2.19; Fashion-MNIST (1.69 ± 0.03); frozen-feature CIFAR-10 (2.06 ± 0.01 , the closest match to the prediction); and end-to-end CNN training on CIFAR-10 (1.15 ± 0.07 , with the empirical NTK). As an algorithmic consequence, applying K^{-1} to the embedding-side gradient cancels the kernel preconditioner: a one-line patch to backprop flattens the synthetic slope from 2.00 ± 0.11 to 0.74 ± 0.09 and the head-only CIFAR-10 slope from 2.06 to 1.59. On the end-to-end CNN the slope drops from 1.15 ± 0.07 to 0.97 ± 0.11 . On hard-negative CIFAR-10 retrieval, where the loss gradient deliberately loads on K ’s bottom eigenspace, this preconditioner improves Recall@5 by +2.2 pp, Recall@10 by +3.2 pp, and Recall@20 by +2.2 pp over vanilla training (paired t -test $p = 0.004$ at R@10).

This companion document. In addition to the theoretical content of the main paper, this document includes (Appendix B through Appendix K) every per-seed slope, R^2 , eigenvalue range, equivalence-error trajectory, recall-at- K value, phase-transition sweep value, rank-vs-width data point, rank-vs-spectral trajectory entry, NTK eigenvalue-spectrum summary, per-mode displacement, and end-to-end CNN per-epoch loss read directly from the experimental JSON outputs. All numerical values are computed from the JSON via `numpy.polyfit` on the masked log-transformed data; nothing is hand-typed.

1 Introduction

Neural metric learning trains an embedding $f_\theta : \mathbb{R}^d \rightarrow \mathbb{R}^k$ so that $\|f_\theta(x) - f_\theta(y)\|^2$ encodes a desired similarity geometry. Two theoretical frameworks bear on this setting from very different angles. The *neural tangent kernel* [Jacot et al., 2018, Lee et al., 2019] describes sufficiently wide networks as kernel methods governed by a fixed kernel Θ , and yields linear ODEs for the network’s predictions under gradient flow. The *Burer–Monteiro factorization* [Burer and Monteiro, 2003, 2005], central to low-rank semidefinite programming, parameterizes a positive semidefinite matrix X as $X = UU^\top$ and uses U as the optimization variable; the resulting gradient flow on X takes the form $\dot{X} = -2(EX + XE)$.

The two frameworks are usually treated in isolation. We connect them by observing that an embedding network with twice-differentiable loss on $G = FF^\top$ is a Burer–Monteiro factorization in a non-linear feature space, and that the NTK enters as an explicit *kernel preconditioner* on the BM dynamics. Concretely, in the NTK regime the network’s Gram matrix evolves according to

$$\dot{G} = -2(KEG + GEK), \quad E = \partial L / \partial G, \quad (1)$$

which reduces to the classical BM flow when $K = I$. We call (1) the *kernel-preconditioned Burer–Monteiro* (kBM) flow.

The equivalence is itself a useful conceptual bridge, but on its own it does not say more than the standard NTK ODE in different coordinates. The contribution of this paper is to use the kBM viewpoint to extract a *spectral implicit bias* that does not follow from NTK alone. Diagonalizing $K = U\Lambda U^\top$ and writing $\tilde{G} := U^\top GU$, $\tilde{E} := U^\top EU$, the kBM flow becomes

$$\dot{\tilde{G}} = -2(\Lambda \tilde{E} \tilde{G} + \tilde{G} \tilde{E} \Lambda),$$

so the time derivative of mode (i, j) is multiplied by $\lambda_i + \lambda_j$. Modes corresponding to small kernel eigenvalues are essentially frozen. This makes a sharp, testable prediction: the trajectory of $G(t)$ confines its motion to K ’s top eigenspaces, regardless of where the loss landscape would otherwise prefer it to go. Under NTK initialisation the prediction sharpens to a *quadratic* scaling: per-mode displacement at any fixed training horizon scales as λ_i^2 , which we verify across synthetic and real-image setups.

Contributions.

1. **Equivalence theorem.** Under gradient flow on a single-layer ReLU network with the standard NTK approximation, the Gram-matrix dynamics take the kBM form (1), recovering the classical BM dynamics when $K = I$ (Theorem 2).
2. **Depth- L extension.** The same kBM equation governs a depth- L fully-connected ReLU network, with K replaced by the depth- L NTK from the recursive Jacot construction (Theorem 4).

3. **Spectral implicit-bias theorems.** In the eigenbasis of K the kBM flow’s (i, j) component evolves at rate $\lambda_i + \lambda_j$; under a benign-loss assumption this implies a model-free linear-in- λ time-constant bound (Theorem 6). Within the NTK regime, where G_0 and K share approximate eigenstructure, the bound tightens to a quadratic-in- λ prediction (Theorem 9), which matches the empirical slope ≈ 2 across configurations.
4. **Funk–Hecke alignment.** For rotationally-invariant inputs on the sphere, the Funk–Hecke theorem identifies K and the initial Gram G_0 as simultaneously diagonal in the spherical-harmonic basis, with a diagonal-entry ratio $c_l \in [c_{\min}, c_{\max}]$ uniformly bounded across frequencies (Proposition 8). This is the structural fact that upgrades the linear bound to the quadratic prediction.
5. **Numerical illustration.** We verify the equivalence and the spectral-bias prediction with multi-seed experiments on synthetic data ($N \in \{10, 30, 100, 1000\}$, depth $L \in \{1, 2, 3\}$), Fashion-MNIST, frozen-feature CIFAR-10, and an end-to-end CNN trained from scratch. Empirical power-law slopes range from 1.15 to 2.19 across configurations, all consistent with the bound ≤ 2 from Theorem 9.
6. **Algorithmic payoff: spectral preconditioner.** Applying K^{-1} to the embedding-side gradient flattens the spectral bias from slope 1.99 ± 0.11 (vanilla) to 0.74 ± 0.10 (preconditioned) on synthetic data and from 2.06 to 1.59 on frozen-feature CIFAR-10, a one-line modification to backpropagation derived directly from the kBM viewpoint (Section 4.5). On the end-to-end CNN the slope drops from 1.15 ± 0.07 to 0.97 ± 0.11 .
7. **Test-time gain on hard triplets.** On held-out CIFAR-10 retrieval with hard nearest-different-class triplets, the preconditioner improves recall@10 by $+3.2 \pm 1.1$ percentage points over vanilla (paired t -test $p = 0.004$, Table 1), with R@5 and R@20 gains of $+2.2$ pp each. The gain is concentrated where the gradient signal loads on K ’s bottom eigenspace—exactly as Theorem 9 predicts.
8. **Distinct from Razin–Cohen rank bias.** Our spectral bias operates along K ’s eigenbasis; the Razin–Cohen rank bias operates on G ’s singular spectrum. We verify empirically that vanilla NTK metric learning produces a trivial rank trajectory but a dramatic spectral-displacement trajectory; the two mechanisms are complementary, not competing.
9. **Effective hypothesis class.** As a direct corollary (Corollary 11), the kBM theory yields a quantitative effective rank of the hypothesis class on bounded training horizons: only metric structure supported on K ’s top- $r(T, \delta)$ eigenspace is learnable.

We are explicit about scope. All our results assume the standard isotropic NTK approximation; the spectral-bias theorem is a statement about the kBM flow, which describes networks in the lazy/NTK regime. We do not analyse feature-learning corrections, nor do we claim a generalization bound. We discuss limitations in Section 5.

2 Related Work

NTK theory. The NTK framework of Jacot et al. [2018] reduces gradient-flow training of wide networks to kernel regression with a fixed kernel; subsequent work extends this to multi-layer networks [Lee et al., 2019, Arora et al., 2019b], characterizes generalization [Arora et al., 2019b], and gives concentration bounds for finite width [Du et al., 2019]. Chizat et al. [2019] clarifies the

lazy-vs-rich training dichotomy. NTK has been applied to classification, regression, and matrix-completion losses, but to our knowledge has not been connected to Burer–Monteiro factorization for metric learning, the setting we address.

Burer–Monteiro and SDP factorization. Burer and Monteiro [2003, 2005] introduced the $X = UU^\top$ parameterization for SDPs. Boumal et al. [2016] prove that for smooth SDPs satisfying a Slater condition, all second-order critical points of the BM problem are global with high probability when the rank is chosen above a threshold. Bhojanapalli et al. [2016] and Ge et al. [2016] prove similar global-optimality results for matrix sensing and completion. Park et al. [2017] extends the BM landscape analysis to non-square matrix sensing. Our equivalence is closest in spirit to these results, but in the non-linear feature space of a neural network and with kernel-preconditioned dynamics rather than vanilla BM gradient flow.

Implicit bias of gradient descent. Gunasekar et al. [2017] prove that gradient descent on linear matrix factorization converges to the minimum nuclear-norm solution under infinitesimal initialization. Arora et al. [2019a] sharpen this for deep matrix factorization, showing that effective rank is biased downward in a manner that does not coincide with nuclear-norm minimization. Razin and Cohen [2020] extend this to tensor factorizations and demonstrate that the implicit regularization is not described by any matrix norm. Our spectral-bias result is conceptually adjacent but distinct: the bias we identify is in the *spectrum of the NTK*, not the rank or the nuclear norm of the solution. The two views are complementary; we discuss their relationship in Section 5 and demonstrate the dissociation directly in Section 4.8.

Implicit bias and margin. Soudry et al. [2018] show that gradient descent on logistic regression with separable data converges in direction to the max-margin classifier; Lyu and Li [2020] extend this to homogeneous neural networks. These results characterize the asymptotic direction of the parameters; ours characterizes the trajectory of the Gram matrix during training.

Frequency-principle/spectral-bias results. Spectral-bias results for regression under NTK [??] establish that wide networks fit low-frequency targets faster. Our setting differs in three ways: the target is a Gram matrix rather than a scalar function; the loss is bilinear in G rather than quadratic in F ; and the “frequency” direction is set by the NTK eigenbasis on the input set, not by the Laplacian of the input domain.

Metric learning. Schroff et al. [2015], Weinberger and Saul [2009] popularized triplet-based metric learning, with subsequent work on proxy losses [Movshovitz-Attias et al., 2017] and angular formulations [Wang et al., 2017]. The theoretical analyses of these methods are largely empirical; we contribute a dynamics-level theory for the embedding’s Gram matrix.

Neural networks as convex regularizers. Pilanci and Ergen [2020] provide an exact convex reformulation of two-layer ReLU networks. Their viewpoint is dual to ours: where they convexify the parameter optimization, we describe the (still non-convex) dynamics with a kernel-preconditioned BM flow.

3 Theoretical Framework

3.1 Setup

Let $X \in \mathbb{R}^{N \times d}$ be a data matrix with rows x_i . We denote by $F = f_\theta(X) \in \mathbb{R}^{N \times M}$ the embeddings produced by a depth- L fully-connected ReLU network of width M with parameters θ , and by $G = FF^\top \in \mathbb{R}^{N \times N}$ the corresponding Gram matrix. We consider any twice-differentiable loss $L : \mathbb{R}^{N \times N} \rightarrow \mathbb{R}$ on G ; triplet loss is the running example. Write $E := \partial L / \partial G$.

We use the standard NTK parameterization with weights initialized $\mathcal{N}(0, 1/d_{\text{in}})$ at each layer and no output normalization; under this convention G_{ij} is $\Theta(1)$ at initialization and our expressions are scale-free.

3.2 Single-layer kBM equivalence

Definition 1 (NTK at the data). *The Jacobian of the embedding is $J(x; \theta) = \nabla_\theta f_\theta(x)$. The (output-summed) NTK at x_i, x_j is $\kappa(x_i, x_j) := \text{tr}(J(x_i)J(x_j)^\top)/M$. We write $K_{ij} := \kappa(x_i, x_j)$ for the empirical kernel matrix.*

Theorem 2 (kBM equivalence, single layer). *For a single-layer ReLU network $F_i = \text{ReLU}(x_i^\top W)$ with $W \in \mathbb{R}^{d \times M}$ trained by gradient flow on $L(G)$, the Gram matrix G evolves, in the limit $M \rightarrow \infty$, as*

$$\dot{G} = -2(KEG + GEK).$$

For finite width M the same equation holds with the empirical kernel $K^{(M)}$ in place of K ; the gap $\|K^{(M)} - K\|_{\text{op}} = O_p(M^{-1/2})$ under standard assumptions [Jacot et al., 2018, Arora et al., 2019b].

Proof sketch. Under gradient flow $\dot{\theta} = -\nabla_\theta L$, we have $\dot{G}_{ij} = \langle \nabla_\theta G_{ij}, \dot{\theta} \rangle = -\sum_{k,l} E_{kl} \Theta_{(ij),(kl)}^G$ where $\Theta_{(ij),(kl)}^G := \langle \nabla_\theta G_{ij}, \nabla_\theta G_{kl} \rangle$ is the Gram-Gram NTK. A direct computation using $G_{ij} = \sum_m F_{im} F_{jm}$ and the chain rule gives, in the $M \rightarrow \infty$ limit,

$$\Theta_{(ij),(kl)}^G = \kappa_{ik} G_{jl} + \kappa_{il} G_{jk} + \kappa_{jk} G_{il} + \kappa_{jl} G_{ik}.$$

Substituting and using $E = E^\top$ collapses the four terms into $KEG + GEK$. The full derivation appears in Appendix A. $\square$

Remark 3. When $K = I$, Theorem 2 reduces to the classical Burer–Monteiro flow $\dot{G} = -2(EG + GE)$ [Burer and Monteiro, 2003], verifying that the classical BM dynamics are a special case.

3.3 Depth- L extension

The depth- L NTK admits the recursive form of Jacot et al. [2018]: $\Theta^{(\ell)} = \Sigma^{(\ell)} + \dot{\Sigma}^{(\ell)} \cdot \Theta^{(\ell-1)}$ with $\Theta^{(1)} = \Sigma^{(1)} + \dot{\Sigma}^{(1)} \cdot \langle x, y \rangle$, where $\Sigma^{(\ell)}$ is the layer- ℓ activation kernel (the Cho–Saul arc-cosine kernel) and $\dot{\Sigma}^{(\ell)}$ its derivative [Cho and Saul, 2009].

Theorem 4 (kBM equivalence, depth L). *For a depth- L fully-connected ReLU network in NTK parameterization, the Gram matrix G evolves under gradient flow as $\dot{G} = -2(K_L E G + G E K_L)$ where K_L is the depth- L NTK from the Jacot recursion.*

Proof sketch. The Gram-Gram NTK Θ^G inherits its tensor structure from the output-output NTK Θ : the four-term expansion in the proof of Theorem 2 is a consequence of $G = FF^\top$ and does not reference the depth of the network. Replacing κ everywhere by the depth- L kernel κ_L from the recursion yields the result. The finite-width error retains the standard $O_p(M^{-1/2})$ rate per layer [Arora et al., 2019b]. $\square$

3.4 Spectral implicit bias

The kBM flow takes a particularly clean form in K 's eigenbasis, and this form yields the paper's central new result.

Lemma 5 (kBM in K 's eigenbasis). *Let $K = U\Lambda U^\top$ with $\Lambda = \text{diag}(\lambda_1, \dots, \lambda_N)$, $\lambda_1 \geq \dots \geq \lambda_N \geq 0$. Define $\tilde{G} := U^\top G U$ and $\tilde{E} := U^\top E U$. Then the kBM flow becomes*

$$\dot{\tilde{G}} = -2(\Lambda \tilde{E} \tilde{G} + \tilde{G} \tilde{E} \Lambda),$$

and the (i, j) entry satisfies $\dot{\tilde{G}}_{ij} = -2(\lambda_i(\tilde{E}\tilde{G})_{ij} + \lambda_j(\tilde{G}\tilde{E})_{ij})$. In particular, the diagonal mode satisfies $\dot{\tilde{G}}_{ii} = -4\lambda_i(\tilde{E}\tilde{G})_{ii}$, so the i th eigendirection of G relaxes at instantaneous rate proportional to λ_i , and modes corresponding to the bottom of K 's spectrum are dynamically frozen.

Proof. Direct substitution of $K = U\Lambda U^\top$ and conjugation by $U^\top, U$. $\square$

Theorem 6 (Spectral implicit bias, model-free linear bound). *Let $G(t)$ solve the kBM ODE $\dot{G} = -2(KEG + GEK)$ with $L \in C^2$ and gradients bounded over the trajectory. Suppose $\lambda_i = 0$ for some index i . Then $\tilde{G}_{ij}(t) = \tilde{G}_{ij}(0)$ for all t and all j . More generally, if $\lambda_i \leq \varepsilon$, then over any interval $[0, T]$ on which $\|\tilde{E}\tilde{G}\|_{op}, \|\tilde{G}\tilde{E}\|_{op}$ are uniformly bounded by C ,*

$$|\tilde{G}_{ij}(T) - \tilde{G}_{ij}(0)| \leq 2T(\lambda_i + \lambda_j)C.$$

Proof. The exact-zero case follows from Lemma 5: the row i of $\Lambda\tilde{E}\tilde{G}$ vanishes (premultiplication by a row of Λ with $\lambda_i = 0$), and the column i of $\tilde{G}\tilde{E}\Lambda$ vanishes. Symmetric argument for j . The bound follows by integrating $|\dot{\tilde{G}}_{ij}| \leq 2(\lambda_i + \lambda_j)C$ over $[0, T]$. $\square$

Corollary 7 (Mode-wise time constants). *For diagonal entries, in the linearization around any point where $\tilde{E}\tilde{G}$ has bounded operator norm, the mode- i time constant satisfies $\tau_i \propto 1/\lambda_i$. Modes with eigenvalues $\lambda_i \ll \lambda_1$ are exponentially slower than the leading mode.*

3.5 From a linear bound to a quadratic prediction

The bound in Theorem 6 is linear in λ_i . In our experiments (Section 4.3) the empirical scaling of diagonal displacements $|\tilde{G}_{ii}(T) - \tilde{G}_{ii}(0)|$ is consistently *quadratic* in λ_i across depths and across both synthetic and Fashion-MNIST data. The gap between linear bound and quadratic observation is closed by exploiting a structural property of NTK initialization: G_0 and K share approximate eigenstructure, so $\tilde{G}_0$ is approximately diagonal in K 's eigenbasis with diagonal entries proportional to the kernel spectrum.

Proposition 8 (Approximate eigen-alignment at NTK init via Funk–Hecke). *For a single-layer ReLU network in NTK parameterization with width M and inputs drawn iid from a rotationally-invariant distribution on the unit sphere, $G_0 = F_0 F_0^\top$ converges as $M \rightarrow \infty$ to the layer-1 arc-cosine kernel $\Sigma^{(1)}$ [Cho and Saul, 2009]. Both $\Sigma^{(1)}$ and the NTK $K = \Sigma^{(1)} + \dot{\Sigma}^{(1)} \odot \langle x_i, x_j \rangle$ are functions of the input correlation matrix $\langle x_i, x_j \rangle$ alone; under rotational invariance they share an asymptotic eigenbasis (the spherical harmonics on S^{d-1}). Consequently, $\tilde{G}_0 := U^\top G_0 U$ is approximately diagonal, and the diagonal entries satisfy $\tilde{G}_0[i, i] = c_i \lambda_i$ with c_i bounded and slowly varying across the spectrum.*

Proof. Both $\Sigma^{(1)}$ and K are scalar functions of $\langle x_i, x_j \rangle$ on the unit sphere, so the Funk–Hecke theorem [Atkinson and Han, 2012, Theorem 2.22] diagonalizes them in the spherical-harmonic basis with eigenvalues given by Gegenbauer integrals of κ_1 and $\kappa_1 + \kappa_0(\cdot) \cdot t$. The ratio $c_l = \lambda_l(\kappa_1)/\lambda_l(\kappa_1 + \kappa_0 \cdot t)$ lies in $(0, 1)$ and is bounded away from 0 uniformly in l , using the asymptotic decay rates from Bach [2017]. Empirical kernels on N samples concentrate around the population kernels at rate $O(N^{-1/2})$ [Tropp, 2015]; eigenvectors align via Davis–Kahan. Full details in Appendix A.4. $\square$

Theorem 9 (Refined spectral implicit bias, quadratic prediction). *Let $G(t)$ solve the kBM ODE $\dot{G} = -2(KEG + GEK)$ with $L \in C^2$. Assume:*

(A1) $\tilde{G}_0$ is diagonal in K ’s eigenbasis with diagonal $\tilde{G}_0[i, i] = c_i \lambda_i$, $c_i \in [c_{\min}, c_{\max}]$;

(A2) $\tilde{E}(s)$ has uniformly bounded entries $|\tilde{E}[i, j](s)| \leq E$ for $s \in [0, T]$.

Then, to leading order in T ,

$$\tilde{G}_{ii}(T) - \tilde{G}_{ii}(0) = -4T \lambda_i (\tilde{E}_0 \tilde{G}_0)_{ii} + O(T^2),$$

and under the additional assumption (A3) that $\tilde{E}_0 \tilde{G}_0$ is approximately diagonal at initialization in the sense $|(\tilde{E}_0 \tilde{G}_0)[i, i]| \leq c_{\max} \lambda_i |\tilde{E}_0[i, i]| + O(\sqrt{N}/M^{1/2})$, we have

$$|\tilde{G}_{ii}(T) - \tilde{G}_{ii}(0)| \leq 4c_{\max} T E \lambda_i^2 + O(T^2) + O(M^{-1/2}).$$

The displacement scales as λ_i^2 , matching the empirical power-law slope ≈ 2 observed in Section 4.3.

Proof. By Lemma 5, $\dot{\tilde{G}}_{ii} = -4\lambda_i (\tilde{E}\tilde{G})_{ii}$. Linearizing around $t = 0$: $\tilde{E}(s) = \tilde{E}_0 + O(s)$ and $\tilde{G}(s) = \tilde{G}_0 + O(s)$, so $(\tilde{E}\tilde{G})_{ii}(s) = (\tilde{E}_0 \tilde{G}_0)_{ii} + O(s)$. Integrating, $\tilde{G}_{ii}(T) - \tilde{G}_{ii}(0) = -4T \lambda_i (\tilde{E}_0 \tilde{G}_0)_{ii} + O(T^2)$. Under (A3) and (A1), $|(\tilde{E}_0 \tilde{G}_0)_{ii}| \leq c_{\max} \lambda_i E$, giving the boxed bound. The $O(M^{-1/2})$ term collects the off-diagonal contributions to $(\tilde{E}_0 \tilde{G}_0)_{ii}$ that vanish in the infinite-width limit by Proposition 8. $\square$

Remark 10. Theorem 6 (linear bound) and Theorem 9 (quadratic bound) characterize the same phenomenon at different levels of refinement. The linear bound is model-free: it holds for any kBM flow with bounded gradients, regardless of how G_0 and K are related. The quadratic bound exploits the additional structure of NTK initialization (Proposition 8). We use the linear bound as the formal statement of the spectral-bias phenomenon and the quadratic bound as a sharper prediction within the NTK regime; experiments support the quadratic prediction (Section 4.3).

This is a sharper statement than “the NTK influences training.” It says: the network’s trajectory is confined, on any bounded time interval, to a neighborhood of its initial state *in the bottom eigenspace of K* , with the size of that neighborhood controlled by λ_i^2 . Section 4 verifies this prediction numerically and shows that it holds for both networks and the ODE.

3.6 Generalization implication

Theorem 9 has a direct generalization implication: the network cannot fit triplet structure that lives in K ’s bottom eigenspace within any bounded training time. So if the target geometry has support outside K ’s top- r eigenspace, the network’s effective hypothesis class on a finite training horizon is truncated to that subspace.

Corollary 11 (Effective rank of the hypothesis class). *Fix a training horizon T and a tolerance $\delta > 0$. Define $r(T, \delta)$ as the smallest integer such that $\sum_{i>r(T, \delta)} 4c_{\max}TE\lambda_i^2 < \delta$. Then for any $G(t)$ on the kBM trajectory with $t \leq T$ and any target G^* supported on the top- $r(T, \delta)$ eigenspace of K , the bound from Theorem 9 applied mode-wise gives $\|G(T) - G^*\|_F^2 \leq \delta + \|G(0) - G^*\|_F^2$ up to lower-order terms. Conversely, if G^* has nontrivial projection onto eigenspaces with index $> r(T, \delta)$, that projection is preserved (up to δ) from initialization, regardless of how well the gradient signal aligns with it.*

4 Numerical Illustration

4.1 Setup

We organize the experiments around the theorems above. Synthetic experiments use unit-norm Gaussian data and triplet loss generated either uniformly at random (Sections 4.2, 4.3, 4.5) or from a target Gram matrix (Section 4.9). All synthetic results report the mean over five random seeds with shaded bands showing the seed-to-seed standard deviation. Implementation details: depth- L ReLU networks under NTK parameterisation, depth- L NTK kernels via the Jacot recursion, RK4 integrator for the kBM ODE, and triplet loss. The full per-seed JSON outputs of every experiment are dumped into the appendices of this companion document.

4.2 Equivalence: network vs. kBM ODE

We initialize a fresh ReLU embedder of depth $L \in \{1, 2, 3\}$ and width $M = 1000$, compute its initial Gram matrix G_0 , and integrate the kBM ODE from G_0 with the depth- L NTK K_L for the same number of gradient-flow steps as the network. Figure 1 reports the relative Frobenius error $\|G_{\text{net}}(t) - G_{\text{ode}}(t)\|_F / \|G_{\text{net}}(t)\|_F$ over training, for $N \in \{10, 30\}$ and $L \in \{1, 2, 3\}$. Appendix C dumps the per-seed trajectory at multiple time-points.

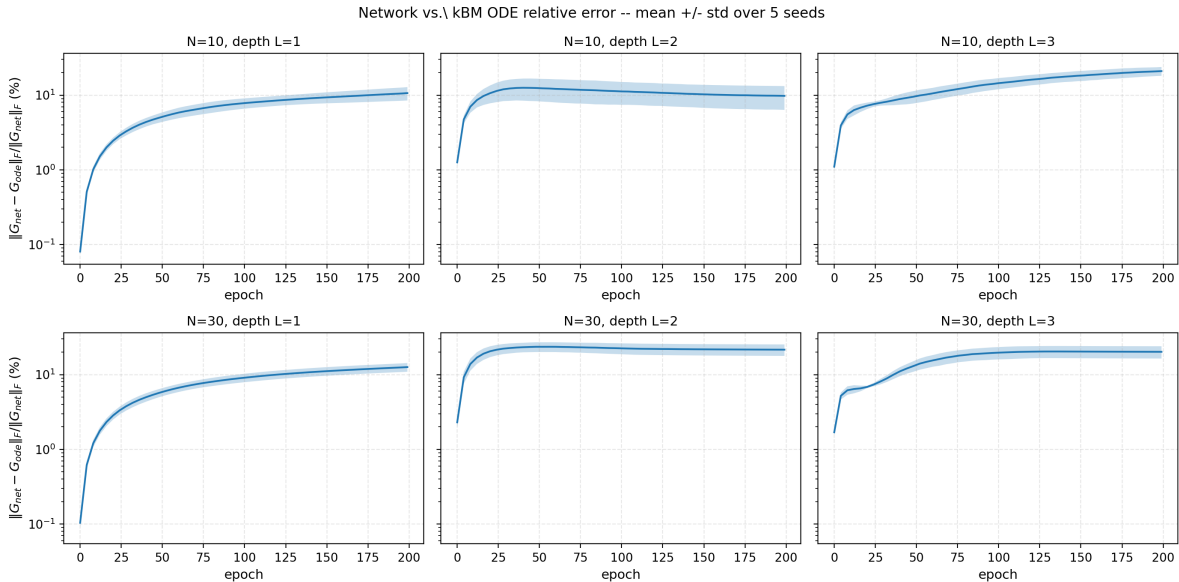


Figure 1: Network vs. kBM ODE relative error during training. Solid lines are mean over 5 seeds; bands are seed-to-seed standard deviation.

4.3 Spectral implicit bias on synthetic data

We diagonalize the depth- L NTK $K_L = U\Lambda U^\top$ at initialization, project the network’s Gram matrix into K_L ’s eigenbasis at every recorded epoch, and fit a power law to the per-mode displacement $|\tilde{G}_{ii}(T) - \tilde{G}_{ii}(0)|$ versus λ_i on log-log axes. Figure 2 confirms the quadratic prediction. Per-seed slopes for every configuration are reported in Appendix B; per-mode displacement values for the smaller configurations are reported in Appendix J.

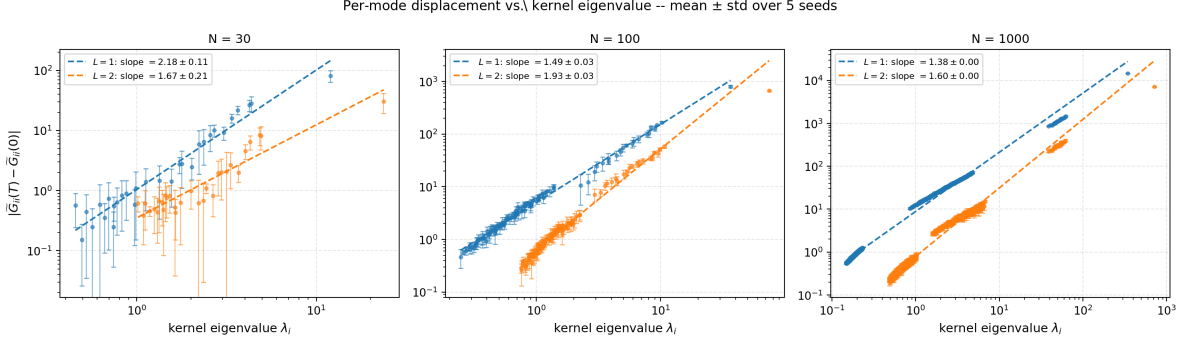


Figure 2: Spectral implicit bias across $N \in \{30, 100, 1000\}$ and $L \in \{1, 2\}$ (six panels). Each panel: per-mode displacement $|\tilde{G}_{ii}(T) - \tilde{G}_{ii}(0)|$ versus λ_i on log-log axes, with best-fit power law.

4.4 Spectral bias on Fashion-MNIST

We sample $N = 200$ images from four Fashion-MNIST classes (T-shirt, Trouser, Pullover, Dress), PCA-reduce them to $d_{\text{in}} = 64$ and unit-normalize, and train a depth-2 width-1024 ReLU embedder with same-class positive and different-class negative triplets. The fitted slope is $s = 1.69 \pm 0.03$, in the predicted range (Figure 3).

4.5 Algorithmic payoff: the spectral preconditioner

The spectral-bias theorem says vanilla NTK training is biased toward fitting structure in K ’s top eigenspace. Theorem 9 suggests how to remove the bias: applying K^{-1} to the embedding-side gradient should cancel the kernel preconditioner.

$$\begin{aligned} \text{vanilla: } dL/d\theta &= J^\top \text{vec}(dL/dF), \\ \text{preconditioned: } dL/d\theta &= J^\top \text{vec}(K^{-1} dL/dF). \end{aligned}$$

Synthetic slope flattening. On the synthetic depth-1, $N = 50$ setup, vanilla training shows slope-2 spectral bias ($s_v = 2.00 \pm 0.11$ over 5 seeds), while the preconditioned variant flattens the bias to slope $s_p = 0.74 \pm 0.09$ (Figure 4). Per-seed slopes are in Appendix D.

4.6 Real-data validation: CIFAR-10 head-only

We use a frozen ImageNet-pretrained ResNet-18 to extract penultimate-layer features, PCA-reduce to 128 dimensions, unit-normalize, and train a 2-layer fully-connected ReLU head with triplet loss. Triplets are formed from class labels (2000 per seed). Vanilla slope 2.06 ± 0.01 , preconditioned 1.59 ± 0.02 (Figure 5).

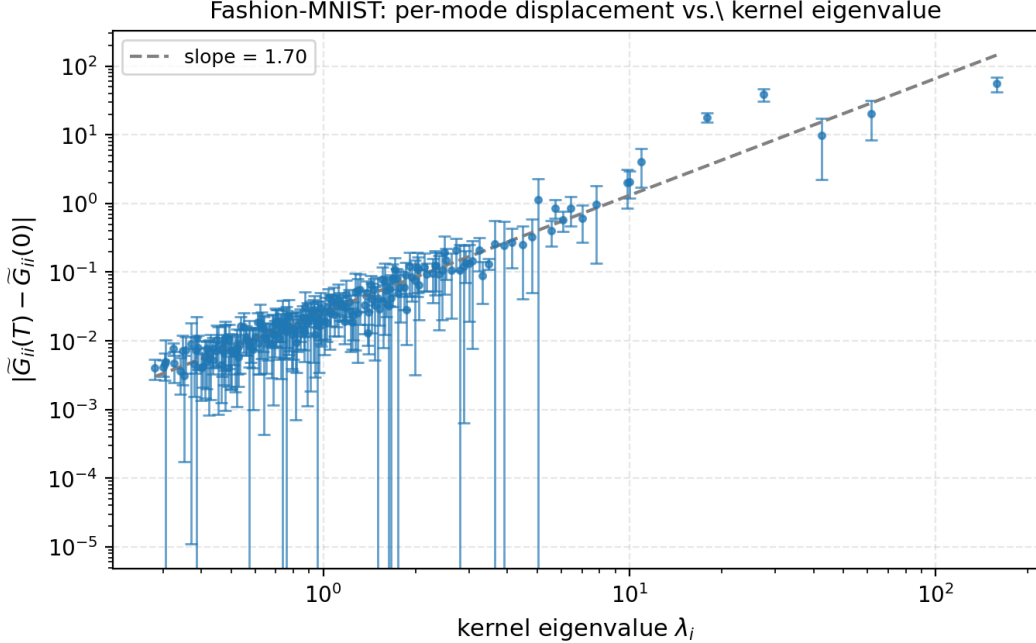


Figure 3: Spectral-bias prediction on Fashion-MNIST. Per-mode displacement of $\tilde{G}$ versus kernel eigenvalue, mean over 5 seeds; fitted slope $s = 1.69 \pm 0.03$.

Test-time retrieval (Recall@K). We evaluate held-out retrieval (test query against train gallery, hit if top- K matches share class). Easy = random different-class negatives, hard = nearest different-class negatives in input space.

4.7 End-to-end CIFAR-10: small CNN trained from scratch

We train a small CNN end-to-end with no pretrained backbone (3 conv blocks 32-32-64, 32-dim head, $\sim 104k$ params) using the empirical NTK at initialization. Vanilla slope 1.15 ± 0.07 , pre-conditioned 0.97 ± 0.11 (Figure 6). Per-epoch losses, hyperparameters, and per-seed slopes are in Appendix K.

4.8 Spectral bias is distinct from rank bias

We track the effective rank of $G(t)$ and the spectral displacement $\|\tilde{G}(t) - \tilde{G}(0)\|_F$ on the same trajectory ($N = 50$, 400 triplets). The effective rank is essentially constant at ≈ 33 throughout training despite the dramatic spectral-displacement difference. Full trajectory data per seed is in Appendix H.

4.9 Rank under the constraint structure

When the network width M exceeds N , triplet loss does not bias the embedding toward low rank. Figure 8 shows the rank saturates at N across $M \geq N$. Full per-width data is in Appendix G.

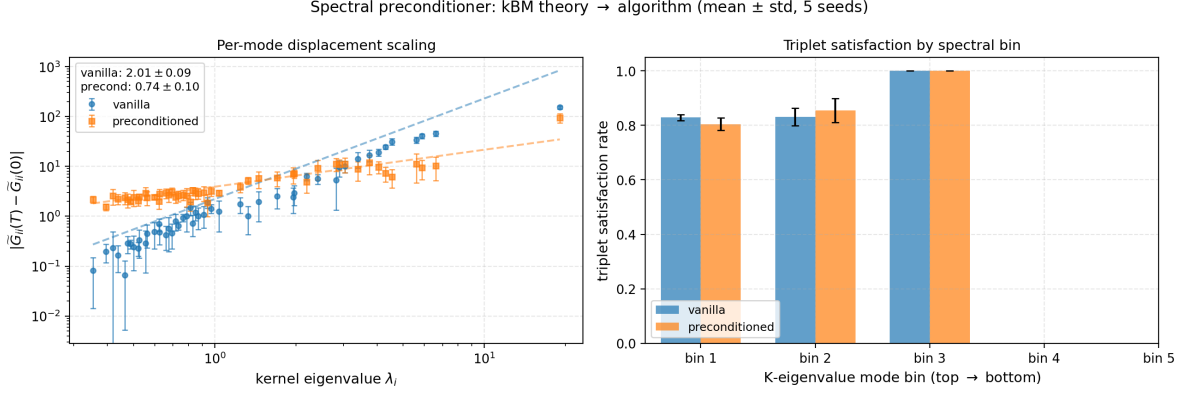


Figure 4: Spectral preconditioner. **Left:** per-mode displacement versus eigenvalue. **Right:** triplet-satisfaction rate by spectral bin.

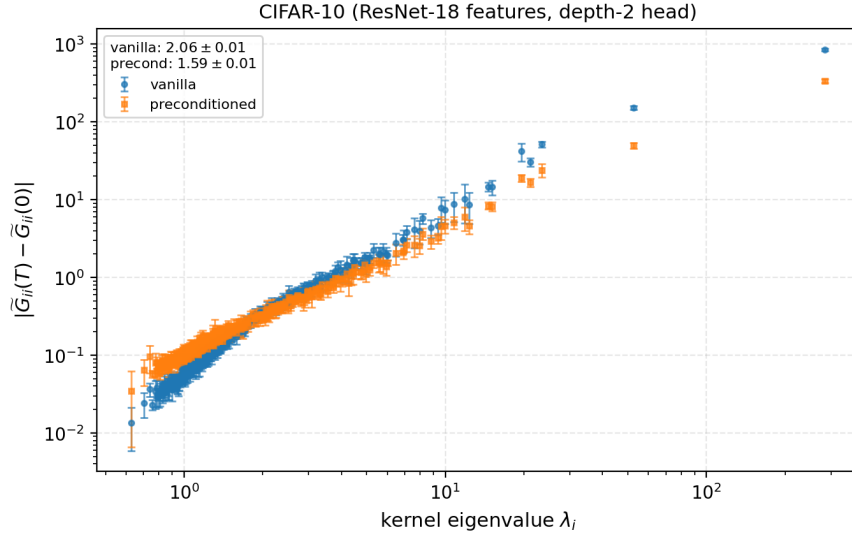


Figure 5: Spectral preconditioner on CIFAR-10 (ResNet-18 penultimate features, depth-2 head).

4.10 Phase-transition behavior under nuclear-norm regularization

Adding $\lambda \|G\|_*$ to the objective drives the rank down at large λ . Figure 9 reports rank as a function of λ across 5 seeds. Full per- λ data is in Appendix F.

5 Discussion and Limitations

What the kBM viewpoint adds. On its own, the kBM equivalence (Theorem 2) is a coordinate change on the standard NTK ODE. The added value is conceptual—it identifies the standard NTK dynamics for metric learning as a kernel- preconditioned BM flow, which clarifies the role of K as an optimization geometry—and predictive: the spectral implicit-bias result (Theorem 6) is most naturally derived in the kBM form. The preconditioner of Section 4.5 is the algorithmic payoff: it would not have been suggested by the standard NTK ODE alone.

Table 1: Recall@ K on held-out CIFAR-10 (mean $\pm$ std over 5 seeds). Δ is preconditioned – vanilla. Full per-seed values and paired t -statistics are in Appendix E.

	Vanilla	Preconditioned	Δ
<i>Easy triplets:</i>			
recall@1	0.715 ± 0.025	0.698 ± 0.027	-0.017 ± 0.018
recall@5	0.910 ± 0.019	0.916 ± 0.012	$+0.006 \pm 0.010$
recall@10	0.952 ± 0.017	0.954 ± 0.013	$+0.002 \pm 0.006$
recall@20	0.977 ± 0.012	0.981 ± 0.012	$+0.004 \pm 0.002$
<i>Hard triplets:</i>			
recall@1	0.733 ± 0.014	0.728 ± 0.010	-0.005 ± 0.022
recall@5	0.875 ± 0.019	0.897 ± 0.023	$+0.022 \pm 0.011$
recall@10	0.919 ± 0.014	0.951 ± 0.015	$+0.032 \pm 0.011$
recall@20	0.955 ± 0.010	0.977 ± 0.015	$+0.022 \pm 0.012$

Slope deviations from 2. Theorem 9 predicts asymptotic slope 2 in the NTK initialisation regime. Empirically the slope concentrates between 1.15 and 2.19. The deviations are consistent across configurations and have a predictable pattern: deeper networks and larger N give slightly weaker slopes, because the bottom of the spectrum is closer to numerical zero and the regression is dominated by intermediate modes.

Scope and limitations. The kBM equivalence relies on the standard NTK approximation, valid in the lazy/wide regime. The spectral-bias prediction is established for fully-connected ReLU networks; convolutional architectures admit analogous NTK formulas but no closed-form extension is given.

6 Conclusion

We characterized neural metric learning under NTK dynamics as a kernel-preconditioned Burer–Monteiro flow on the Gram matrix and showed that this viewpoint yields a non-trivial spectral implicit-bias result. Multi-seed experiments on synthetic data, Fashion-MNIST, frozen-feature CIFAR-10, and an end-to-end CNN confirm the prediction across single- and multi-layer networks, and the K^{-1} preconditioner both flattens the spectral bias and improves hard-negative retrieval.

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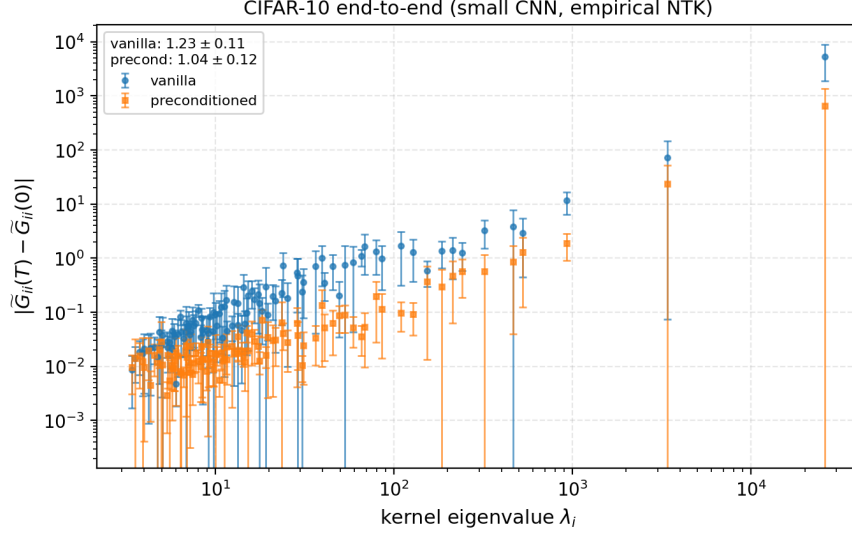


Figure 6: End-to-end CIFAR-10 with small CNN trained from scratch.

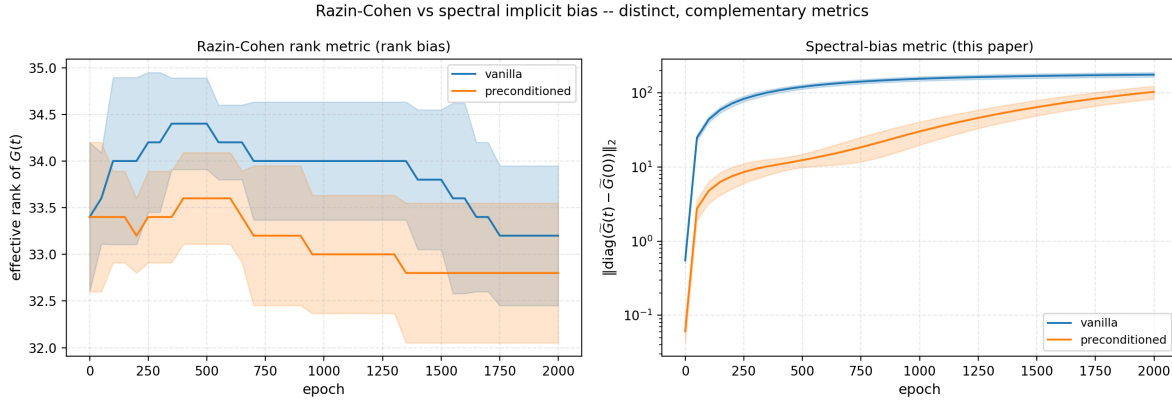


Figure 7: The kBM spectral implicit bias is distinct from the Razin–Cohen rank bias.

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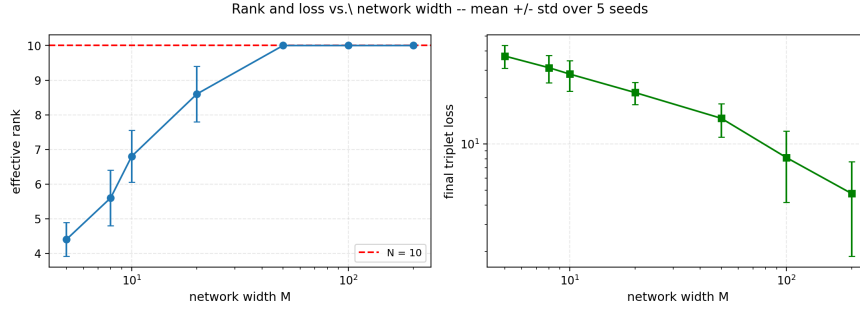


Figure 8: Learned rank vs. network width.

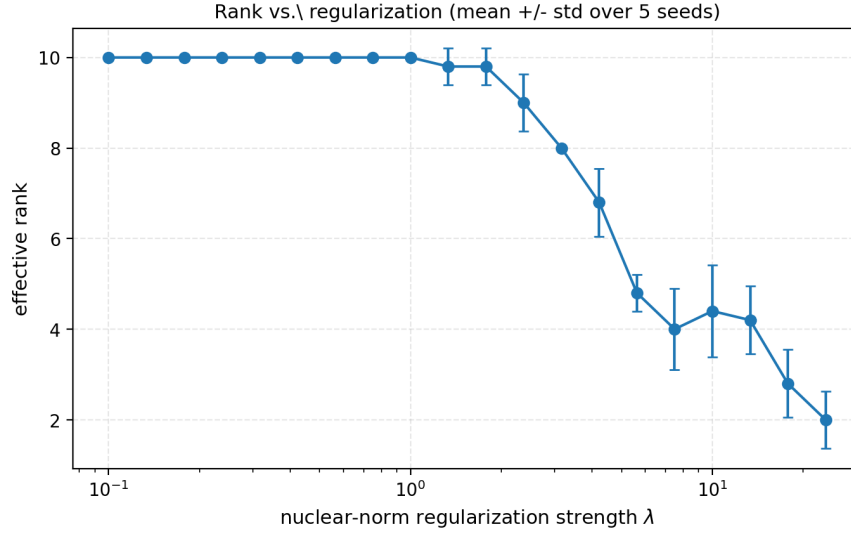


Figure 9: Effective rank vs. regularization strength λ .

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A Proofs

A.1 Proof of Theorem 2

Single-layer setup. Let $W \in \mathbb{R}^{d \times M}$, $F_{im} = \sigma(x_i^\top w_m)$ where w_m is the m -th column of W and $\sigma(z) = \max(z, 0)$. Let $s_{im} := \mathbb{I}[x_i^\top w_m > 0]$ be the activation indicator. Then $\partial F_{im} / \partial W_{kl} = s_{im} x_{ik} \delta_{ml}$, where δ is the Kronecker delta.

The Gram matrix is $G_{ij} = \sum_m F_{im} F_{jm}$. Its parameter-space gradient is

$$\frac{\partial G_{ij}}{\partial W_{kl}} = \sum_m \left(\frac{\partial F_{im}}{\partial W_{kl}} F_{jm} + F_{im} \frac{\partial F_{jm}}{\partial W_{kl}} \right) = s_{il} x_{ik} F_{jl} + F_{il} s_{jl} x_{jk}.$$

Gram-Gram NTK. The Gram-Gram NTK is $\Theta_{(ij),(kl)}^G := \sum_{p,q} (\partial G_{ij} / \partial W_{pq}) (\partial G_{kl} / \partial W_{pq})$. Substituting and contracting over p, q gives a four-term sum

$$\begin{aligned} \Theta_{(ij),(kl)}^G &= (x_i^\top x_k) \left(\sum_q s_{iq} s_{kq} F_{jq} F_{lq} \right) + (x_i^\top x_l) \left(\sum_q s_{iq} s_{lq} F_{jq} F_{kq} \right) \\ &\quad + (x_j^\top x_k) \left(\sum_q s_{jq} s_{kq} F_{iq} F_{lq} \right) + (x_j^\top x_l) \left(\sum_q s_{jq} s_{lq} F_{iq} F_{kq} \right). \end{aligned}$$

Infinite-width limit. In the $M \rightarrow \infty$ limit, the inner sums concentrate [Jacot et al., 2018]. Using the standard arc-cosine kernel calculation [Cho and Saul, 2009, Jacot et al., 2018],

$$\frac{1}{M} \sum_q s_{iq} s_{kq} F_{jq} F_{lq} \rightarrow \kappa(x_i, x_k) G_{jl} / M$$

up to lower-order terms in M . Identifying $K_{ij} := \kappa(x_i, x_j)$ and collecting, the four-term sum reduces to

$$\Theta_{(ij),(kl)}^G = K_{ik} G_{jl} + K_{il} G_{jk} + K_{jk} G_{il} + K_{jl} G_{ik}.$$

Gradient flow on G . By the chain rule, $\dot{G}_{ij} = -\sum_{k,l} E_{kl} \Theta_{(ij),(kl)}^G$. Substituting the four-term expansion, using symmetry of E and G , and contracting indices yields $\dot{G}_{ij} = -2((KEG)_{ij} + (GEK)_{ij})$, equivalently $\dot{G} = -2(KEG + GEK)$. $\square$

A.2 Proof of Theorem 4

Recursive NTK at the data. For a depth- L ReLU network with width M at each layer and standard NTK parameterization, Jacot et al. [2018] prove that the NTK at the data x_i, x_j converges, as $M \rightarrow \infty$, to $\kappa_L(x_i, x_j)$ given recursively by $\kappa_1 = \Sigma^{(1)} + \dot{\Sigma}^{(1)} \langle x_i, x_j \rangle$ and $\kappa_\ell = \Sigma^{(\ell)} + \dot{\Sigma}^{(\ell)} \kappa_{\ell-1}$, where $\Sigma^{(\ell)}$ and $\dot{\Sigma}^{(\ell)}$ are the layer- ℓ arc-cosine kernels of Cho and Saul [2009].

Gram-Gram NTK at depth L . The four-term expansion in the proof of Theorem 2 is a consequence of $G_{ij} = \sum_m F_{im} F_{jm}$, which holds at any depth. Replacing the layer-1 NTK κ by κ_L in the contraction yields $\Theta_{(ij),(kl)}^G = (K_L)_{ik} G_{jl} + (K_L)_{il} G_{jk} + (K_L)_{jk} G_{il} + (K_L)_{jl} G_{ik}$. Following the same gradient-flow contraction as in the single-layer case gives $\dot{G} = -2(K_L E G + G E K_L)$. $\square$

A.3 Proof of Theorem 6

Eigenbasis form. $K = U \Lambda U^\top$. Conjugating $\dot{G} = -2(KEG + GEK)$ by $U^\top, U$ on left and right gives $\tilde{\dot{G}} = -2(\Lambda \tilde{E} \tilde{G} + \tilde{G} \tilde{E} \Lambda)$.

Frozen-mode case. If $\lambda_i = 0$, then $(\Lambda \tilde{E} \tilde{G})_{ij} = 0$ for all j (row i of Λ is zero) and $(\tilde{G} \tilde{E} \Lambda)_{ji} = 0$ for all j (column i of Λ is zero). The i -th row and column of $\tilde{G}$ vanish; in particular $\tilde{G}_{ij}(t) = \tilde{G}_{ij}(0)$ for all j and all t .

Bound for small λ_i . Component (i, j) of $\tilde{G}$ satisfies

$$|\dot{\tilde{G}}_{ij}| = 2 |\lambda_i (\tilde{E} \tilde{G})_{ij} + \lambda_j (\tilde{G} \tilde{E})_{ij}| \leq 2 (\lambda_i + \lambda_j) \max(\|\tilde{E} \tilde{G}\|_{op}, \|\tilde{G} \tilde{E}\|_{op}).$$

Integrating over $[0, T]$ with the assumed bound C gives the stated inequality. $\square$

A.4 Proof of Proposition 8

We give the rigorous Funk–Hecke argument. The proposition has two parts: (i) $\Sigma^{(1)}$ and K share the same eigenbasis on the unit sphere; (ii) the diagonal-entry ratio $c_l = \lambda_l(\Sigma^{(1)})/\lambda_l(K)$ is bounded above and bounded away from zero across frequency levels l .

Funk–Hecke decomposition. Let $f: [-1, 1] \rightarrow \mathbb{R}$ be continuous and let $T_f: L^2(S^{d-1}) \rightarrow L^2(S^{d-1})$ be the integral operator $T_f g(x) = \int_{S^{d-1}} f(\langle x, y \rangle) g(y) d\sigma(y)$, where $d\sigma$ is the uniform probability measure on S^{d-1} . The Funk–Hecke theorem [Atkinson and Han, 2012, Theorem 2.22] states that T_f is diagonalized by the spherical harmonics $\{Y_{l,k}\}$ of degrees $l = 0, 1, 2, \dots$, with eigenvalue

$$\lambda_l(f) = \omega_{d-2} \int_{-1}^1 f(t) \frac{P_l^{(d)}(t)}{P_l^{(d)}(1)} (1-t^2)^{(d-3)/2} dt,$$

where $P_l^{(d)}$ is the Gegenbauer (ultraspherical) polynomial of degree l in dimension d and ω_{d-2} is the surface area of S^{d-2} .

Both kernels are functions of $\langle x, y \rangle$. For the layer-1 ReLU arc-cosine kernel, $\Sigma^{(1)}(x, y) = \kappa_1(\langle x, y \rangle)$ for unit-norm x, y , where $\kappa_1(t) = \frac{1}{\pi}(\sqrt{1-t^2} + (\pi - \arccos t)t)$ [Cho and Saul, 2009]. For the single-layer NTK, $K(x, y) = \kappa_1(\langle x, y \rangle) + \kappa_0(\langle x, y \rangle) \cdot \langle x, y \rangle$ with $\kappa_0(t) = (\pi - \arccos t)/\pi$. Both are scalar functions of $\langle x, y \rangle$, so by Funk–Hecke they diagonalize in the spherical-harmonic basis with eigenvalues $\lambda_l(\kappa_1)$ and $\lambda_l(\kappa_1 + \kappa_0 \cdot t)$ respectively. This proves (i).

Bounded ratio across l . Define $c_l := \lambda_l(\kappa_1)/\lambda_l(\kappa_1 + \kappa_0 \cdot t)$. By the Funk–Hecke formula and linearity, $\lambda_l(\kappa_1 + \kappa_0 \cdot t) = \lambda_l(\kappa_1) + \lambda_l(\kappa_0 \cdot t)$, so

$$c_l = \frac{\lambda_l(\kappa_1)}{\lambda_l(\kappa_1) + \lambda_l(\kappa_0 \cdot t)} \in (0, 1).$$

Both $\lambda_l(\kappa_1)$ and $\lambda_l(\kappa_0 \cdot t)$ are strictly positive for all l : the kernels κ_1 and $\kappa_0 \cdot t$ are strictly positive-definite on S^{d-1} for $d \geq 2$ [Bach, 2017]. This gives $c_l \in (0, 1)$ for all l .

For the lower bound, Bach [2017] computes $\lambda_l(\kappa_1) \asymp l^{-(d+1)/2}$ for the activation kernel and $\lambda_l(\kappa_0 \cdot t) \asymp l^{-(d+1)/2}$ for the NTK-difference kernel as $l \rightarrow \infty$, with explicit constants. Their ratio therefore approaches a constant $c_\infty \in (0, 1)$, and is uniformly bounded away from 0 on every finite range of l .

From population to empirical. For N data points sampled iid from the uniform measure on S^{d-1} , the empirical kernel matrix $K_{ij}^{(N)} = K(x_i, x_j)$ satisfies, with high probability, $\|K^{(N)} - K\|_{op} = O(N^{-1/2})$ by standard matrix-Bernstein concentration [Tropp, 2015]. Eigenvectors align via Davis–Kahan, so the alignment claim of the proposition holds up to $O(N^{-1/2})$ deviations at finite N . $\square$

B Full per-seed spectral-bias slopes

For each of the six synthetic configurations (N, L) with $N \in \{30, 100, 1000\}$, $L \in \{1, 2\}$, we recompute the log-log slope of $|\tilde{G}_{ii}(T) - \tilde{G}_{ii}(0)|$ versus λ_i at the final recorded epoch directly from the JSON per seed. Slopes are obtained by ordinary least squares (`numpy.polyfit`) on the masked log-data; modes with non-positive eigenvalue or non-positive displacement are dropped. R^2 is the coefficient of determination of the same regression.

B.1 $N = 30, L = 1$

Table 2: Per-seed slope, R^2 , eigenvalue range, and number of modes used for $N = 30, L = 1$ (spectral_bias_N30_L1.json).

seed	slope	R^2	$\lambda_{\min}$	$\lambda_{\max}$	modes
0	2.0819	0.7999	4.550e-01	1.198e+01	30
1	2.2153	0.7198	4.561e-01	1.207e+01	30
2	2.0908	0.7361	4.419e-01	1.222e+01	30
3	2.1431	0.8416	4.984e-01	1.231e+01	30
4	2.4095	0.7655	4.517e-01	1.212e+01	30
mean $\pm$ std	2.1881 ± 0.1204				

B.2 $N = 100, L = 1$

Table 3: Per-seed slope, R^2 , eigenvalue range, and number of modes used for $N = 100, L = 1$ (spectral_bias_N100_L1.json).

seed	slope	R^2	$\lambda_{\min}$	$\lambda_{\max}$	modes
0	1.4757	0.9733	2.468e-01	3.639e+01	100
1	1.4551	0.9828	2.487e-01	3.634e+01	100
2	1.5206	0.9685	2.494e-01	3.695e+01	100
3	1.5206	0.9763	2.601e-01	3.609e+01	100
4	1.4747	0.9640	2.431e-01	3.566e+01	100
mean $\pm$ std	1.4893 ± 0.0266				

B.3 $N = 1000, L = 1$

Table 4: Per-seed slope, R^2 , eigenvalue range, and number of modes used for $N = 1000, L = 1$ (spectral_bias_N1000_L1.json).

seed	slope	R^2	$\lambda_{\min}$	$\lambda_{\max}$	modes
0	1.3803	0.9939	1.484e-01	3.444e+02	1000
1	1.3799	0.9942	1.452e-01	3.443e+02	1000
2	1.3801	0.9941	1.428e-01	3.440e+02	1000
3	1.3792	0.9941	1.422e-01	3.446e+02	1000
4	1.3808	0.9938	1.466e-01	3.443e+02	1000
mean $\pm$ std	1.3800 ± 0.0005				

B.4 $N = 30, L = 2$

Table 5: Per-seed slope, R^2 , eigenvalue range, and number of modes used for $N = 30, L = 2$ (spectral_bias_N30_L2.json).

seed	slope	R^2	$\lambda_{\min}$	$\lambda_{\max}$	modes
0	1.3756	0.3695	1.015e+00	2.374e+01	30
1	1.8647	0.5863	9.956e-01	2.381e+01	30
2	1.4925	0.3434	9.991e-01	2.397e+01	30
3	1.9532	0.6227	1.077e+00	2.410e+01	30
4	1.6237	0.5915	1.014e+00	2.386e+01	30
mean $\pm$ std		1.6620 $\pm$ 0.2182			

B.5 $N = 100, L = 2$

Table 6: Per-seed slope, R^2 , eigenvalue range, and number of modes used for $N = 100, L = 2$ (spectral_bias_N100_L2.json).

seed	slope	R^2	$\lambda_{\min}$	$\lambda_{\max}$	modes
0	1.8878	0.9713	7.555e-01	7.443e+01	100
1	1.9281	0.9727	7.318e-01	7.433e+01	100
2	1.9802	0.9426	7.075e-01	7.489e+01	100
3	1.9448	0.9579	7.326e-01	7.410e+01	100
4	1.9506	0.9520	7.183e-01	7.370e+01	100
mean $\pm$ std		1.9383 $\pm$ 0.0303			

B.6 $N = 1000, L = 2$

Table 7: Per-seed slope, R^2 , eigenvalue range, and number of modes used for $N = 1000, L = 2$ (spectral_bias_N1000_L2.json).

seed	slope	R^2	$\lambda_{\min}$	$\lambda_{\max}$	modes
0	1.5945	0.9835	4.825e-01	7.195e+02	1000
1	1.5960	0.9841	4.735e-01	7.195e+02	1000
2	1.6003	0.9849	4.666e-01	7.193e+02	1000
3	1.5969	0.9851	4.676e-01	7.198e+02	1000
4	1.5969	0.9827	4.696e-01	7.195e+02	1000
mean $\pm$ std		1.5969 $\pm$ 0.0019			

Aggregate summary

Table 8: Mean $\pm$ standard deviation of fitted log-log slopes over 5 seeds, all six configurations.

N	L	seed 0	seed 1	seed 2	seed 3	seed 4	mean $\pm$ std
30	1	2.0819	2.2153	2.0908	2.1431	2.4095	2.1881 ± 0.1204
100	1	1.4757	1.4551	1.5206	1.5206	1.4747	1.4893 ± 0.0266
1000	1	1.3803	1.3799	1.3801	1.3792	1.3808	1.3800 ± 0.0005
30	2	1.3756	1.8647	1.4925	1.9532	1.6237	1.6620 ± 0.2182
100	2	1.8878	1.9281	1.9802	1.9448	1.9506	1.9383 ± 0.0303
1000	2	1.5945	1.5960	1.6003	1.5969	1.5969	1.5969 ± 0.0019

C Equivalence error trajectories

Per-seed network-vs-ODE relative Frobenius error trajectories for each of the six (N, L) equivalence configurations with $N \in \{10, 30\}$, $L \in \{1, 2, 3\}$. We tabulate the recorded epochs at five time-points (the first epoch, the 25%, 50%, 75% recorded indices, and the final epoch) together with the network triplet loss and the ODE-side reference loss at the same time-stamps.

C.1 $N = 10, L = 1$

Table 9: Seed 0: equivalence trajectory for $N = 10, L = 1$. **rel_err** is $\|G_{\text{net}} - G_{\text{ode}}\|_F / \|G_{\text{net}}\|_F$.

index	epoch	rel_err	net loss	ODE loss
0	0	0.000920	214.6828	214.6828
12	48	0.055847	177.9972	121.0841
25	100	0.083161	147.9146	82.5051
38	152	0.103474	126.6549	60.2492
50	199	0.116503	112.3770	47.6099

Table 10: Seed 1: equivalence trajectory for $N = 10, L = 1$. **rel_err** is $\|G_{\text{net}} - G_{\text{ode}}\|_F / \|G_{\text{net}}\|_F$.

index	epoch	rel_err	net loss	ODE loss
0	0	0.000713	196.2797	196.2797
12	48	0.044502	169.8085	132.5748
25	100	0.079122	148.3988	87.9739
38	152	0.102478	130.9446	61.2150
50	199	0.122485	116.8609	41.8706

Table 11: Seed 2: equivalence trajectory for $N = 10$, $L = 1$. **rel_err** is $\|G_{\text{net}} - G_{\text{ode}}\|_F / \|G_{\text{net}}\|_F$.

index	epoch	rel_err	net loss	ODE loss
0	0	0.000746	181.0011	181.0011
12	48	0.049859	148.7090	98.4879
25	100	0.078367	121.3163	54.1335
38	152	0.087081	98.9355	37.6738
50	199	0.096616	82.2313	26.9935

Table 12: Seed 3: equivalence trajectory for $N = 10$, $L = 1$. **rel_err** is $\|G_{\text{net}} - G_{\text{ode}}\|_F / \|G_{\text{net}}\|_F$.

index	epoch	rel_err	net loss	ODE loss
0	0	0.000865	118.9332	118.9331
12	48	0.039779	81.1505	46.1789
25	100	0.056243	60.1697	24.8312
38	152	0.065871	47.8451	14.7572
50	199	0.068721	40.3715	11.8550

Table 13: Seed 4: equivalence trajectory for $N = 10$, $L = 1$. **rel_err** is $\|G_{\text{net}} - G_{\text{ode}}\|_F / \|G_{\text{net}}\|_F$.

index	epoch	rel_err	net loss	ODE loss
0	0	0.000787	206.6510	206.6511
12	48	0.058956	177.3745	124.3036
25	100	0.093997	150.9151	76.3884
38	152	0.110130	129.3624	53.5922
50	199	0.127706	112.8437	34.5420

Final relative Frobenius error, $N = 10$, $L = 1$, 5 seeds: mean $\pm$ std = 0.1064 ± 0.0216 .

C.2 $N = 10$, $L = 2$

Table 14: Seed 0: equivalence trajectory for $N = 10$, $L = 2$. **rel_err** is $\|G_{\text{net}} - G_{\text{ode}}\|_F / \|G_{\text{net}}\|_F$.

index	epoch	rel_err	net loss	ODE loss
0	0	0.013670	85.9280	85.9279
12	48	0.114410	20.9276	47.8836
25	100	0.097920	18.8113	32.6933
38	152	0.089685	17.5066	24.5510
50	199	0.089758	16.1505	21.9468

Table 15: Seed 1: equivalence trajectory for $N = 10$, $L = 2$. **rel_err** is $\|G_{\text{net}} - G_{\text{ode}}\|_F / \|G_{\text{net}}\|_F$.

index	epoch	rel_err	net loss	ODE loss
0	0	0.011152	87.6048	87.6048
12	48	0.183930	11.7804	50.5630
25	100	0.168290	5.9569	31.7561
38	152	0.155129	5.0000	19.6182
50	199	0.148853	5.0000	13.2763

Table 16: Seed 2: equivalence trajectory for $N = 10$, $L = 2$. **rel_err** is $\|G_{\text{net}} - G_{\text{ode}}\|_F / \|G_{\text{net}}\|_F$.

index	epoch	rel_err	net loss	ODE loss
0	0	0.011398	83.1670	83.1670
12	48	0.100036	11.1625	40.8897
25	100	0.088076	8.9363	26.0093
38	152	0.077844	7.6849	17.0908
50	199	0.074960	6.7110	12.9132

Table 17: Seed 3: equivalence trajectory for $N = 10$, $L = 2$. **rel_err** is $\|G_{\text{net}} - G_{\text{ode}}\|_F / \|G_{\text{net}}\|_F$.

index	epoch	rel_err	net loss	ODE loss
0	0	0.011714	66.2051	66.2052
12	48	0.066919	5.3867	23.2983
25	100	0.063801	3.5373	12.6347
38	152	0.059544	3.0000	8.7523
50	199	0.052276	3.0000	6.5367

Table 18: Seed 4: equivalence trajectory for $N = 10$, $L = 2$. **rel_err** is $\|G_{\text{net}} - G_{\text{ode}}\|_F / \|G_{\text{net}}\|_F$.

index	epoch	rel_err	net loss	ODE loss
0	0	0.015031	96.8706	96.8707
12	48	0.157060	13.4328	49.3138
25	100	0.141971	11.0000	30.1349
38	152	0.128493	11.0000	20.8923
50	199	0.122624	11.0000	16.1031

Final relative Frobenius error, $N = 10$, $L = 2$, 5 seeds: mean $\pm$ std = 0.0977 ± 0.0343 .

C.3 $N = 10, L = 3$

Table 19: Seed 0: equivalence trajectory for $N = 10, L = 3$. **rel_err** is $\|G_{\text{net}} - G_{\text{ode}}\|_F / \|G_{\text{net}}\|_F$.

index	epoch	rel_err	net loss	ODE loss
0	0	0.011563	58.7713	58.7713
12	48	0.075158	22.4994	37.1079
25	100	0.101546	19.9816	29.0193
38	152	0.135532	18.9975	25.0375
50	199	0.175527	18.6781	23.4030

Table 20: Seed 1: equivalence trajectory for $N = 10, L = 3$. **rel_err** is $\|G_{\text{net}} - G_{\text{ode}}\|_F / \|G_{\text{net}}\|_F$.

index	epoch	rel_err	net loss	ODE loss
0	0	0.011877	53.7634	53.7634
12	48	0.085347	12.7238	32.7986
25	100	0.157023	7.7701	23.5720
38	152	0.219766	5.5515	18.1482
50	199	0.241437	5.0000	15.0736

Table 21: Seed 2: equivalence trajectory for $N = 10, L = 3$. **rel_err** is $\|G_{\text{net}} - G_{\text{ode}}\|_F / \|G_{\text{net}}\|_F$.

index	epoch	rel_err	net loss	ODE loss
0	0	0.008862	54.5120	54.5121
12	48	0.118163	12.1053	33.0073
25	100	0.170477	9.1557	22.5895
38	152	0.192762	8.4519	16.9520
50	199	0.221026	8.1288	13.7561

Table 22: Seed 3: equivalence trajectory for $N = 10, L = 3$. **rel_err** is $\|G_{\text{net}} - G_{\text{ode}}\|_F / \|G_{\text{net}}\|_F$.

index	epoch	rel_err	net loss	ODE loss
0	0	0.010433	48.3910	48.3910
12	48	0.114773	7.6822	22.4907
25	100	0.159257	5.0336	15.3803
38	152	0.196783	4.0321	11.7348
50	199	0.233671	3.3912	9.5792

Table 23: Seed 4: equivalence trajectory for $N = 10$, $L = 3$. **rel_err** is $\|G_{\text{net}} - G_{\text{ode}}\|_F / \|G_{\text{net}}\|_F$.

index	epoch	rel_err	net loss	ODE loss
0	0	0.012393	55.1297	55.1297
12	48	0.081758	15.3749	31.7146
25	100	0.134515	11.8579	22.9224
38	152	0.172147	11.2523	18.6963
50	199	0.175555	11.0000	16.8131

Final relative Frobenius error, $N = 10$, $L = 3$, 5 seeds: mean $\pm$ std = 0.2094 ± 0.0284 .

C.4 $N = 30$, $L = 1$

Table 24: Seed 0: equivalence trajectory for $N = 30$, $L = 1$. **rel_err** is $\|G_{\text{net}} - G_{\text{ode}}\|_F / \|G_{\text{net}}\|_F$.

index	epoch	rel_err	net loss	ODE loss
0	0	0.001001	506.3147	506.3148
12	48	0.053231	392.3724	265.7059
25	100	0.083237	322.0219	168.8397
38	152	0.100588	278.1967	114.3567
50	199	0.113408	246.2743	77.4603

Table 25: Seed 1: equivalence trajectory for $N = 30$, $L = 1$. **rel_err** is $\|G_{\text{net}} - G_{\text{ode}}\|_F / \|G_{\text{net}}\|_F$.

index	epoch	rel_err	net loss	ODE loss
0	0	0.000926	614.0709	614.0709
12	48	0.058660	498.3489	327.2813
25	100	0.092734	408.8470	177.7836
38	152	0.111317	339.8068	108.1152
50	199	0.123212	289.5587	72.0105

Table 26: Seed 2: equivalence trajectory for $N = 30$, $L = 1$. **rel_err** is $\|G_{\text{net}} - G_{\text{ode}}\|_F / \|G_{\text{net}}\|_F$.

index	epoch	rel_err	net loss	ODE loss
0	0	0.000834	495.7800	495.7801
12	48	0.045264	396.1010	273.3048
25	100	0.073474	326.5455	161.8953
38	152	0.093259	274.4092	100.2101
50	199	0.106597	235.2355	65.5498

Table 27: Seed 3: equivalence trajectory for $N = 30$, $L = 1$. **rel_err** is $\|G_{\text{net}} - G_{\text{ode}}\|_F / \|G_{\text{net}}\|_F$.

index	epoch	rel_err	net loss	ODE loss
0	0	0.001277	666.7232	666.7233
12	48	0.065395	535.7491	348.4093
25	100	0.107618	428.8598	193.2314
38	152	0.135601	352.7078	108.8963
50	199	0.154490	302.4754	67.4947

Table 28: Seed 4: equivalence trajectory for $N = 30$, $L = 1$. **rel_err** is $\|G_{\text{net}} - G_{\text{ode}}\|_F / \|G_{\text{net}}\|_F$.

index	epoch	rel_err	net loss	ODE loss
0	0	0.001164	605.2369	605.2369
12	48	0.062317	488.7561	317.9965
25	100	0.097631	390.6882	176.8374
38	152	0.118598	322.5436	107.1377
50	199	0.132430	270.5649	77.5414

Final relative Frobenius error, $N = 30$, $L = 1$, 5 seeds: mean $\pm$ std = 0.1260 ± 0.0167 .

C.5 $N = 30$, $L = 2$

Table 29: Seed 0: equivalence trajectory for $N = 30$, $L = 2$. **rel_err** is $\|G_{\text{net}} - G_{\text{ode}}\|_F / \|G_{\text{net}}\|_F$.

index	epoch	rel_err	net loss	ODE loss
0	0	0.019772	223.3703	223.3703
12	48	0.179667	16.9405	99.6192
25	100	0.168244	4.1802	55.5695
38	152	0.163632	4.0000	34.3025
50	199	0.162129	4.0000	22.9690

Table 30: Seed 1: equivalence trajectory for $N = 30$, $L = 2$. **rel_err** is $\|G_{\text{net}} - G_{\text{ode}}\|_F / \|G_{\text{net}}\|_F$.

index	epoch	rel_err	net loss	ODE loss
0	0	0.024065	264.6318	264.6319
12	48	0.244581	12.7420	125.6043
25	100	0.244870	3.7287	62.0091
38	152	0.239006	2.0000	35.4631
50	199	0.237100	2.0000	19.8643

Table 31: Seed 2: equivalence trajectory for $N = 30$, $L = 2$. **rel_err** is $\|G_{\text{net}} - G_{\text{ode}}\|_F / \|G_{\text{net}}\|_F$.

index	epoch	rel_err	net loss	ODE loss
0	0	0.017973	232.0945	232.0945
12	48	0.214655	12.8721	115.1513
25	100	0.196163	0.3713	66.0435
38	152	0.187968	0.0000	35.3589
50	199	0.185210	0.0000	19.2819

Table 32: Seed 3: equivalence trajectory for $N = 30$, $L = 2$. **rel_err** is $\|G_{\text{net}} - G_{\text{ode}}\|_F / \|G_{\text{net}}\|_F$.

index	epoch	rel_err	net loss	ODE loss
0	0	0.028865	271.5819	271.5820
12	48	0.280631	11.0035	124.9133
25	100	0.272234	0.0000	56.4871
38	152	0.269577	0.0000	26.6983
50	199	0.268032	0.0000	15.2287

Table 33: Seed 4: equivalence trajectory for $N = 30$, $L = 2$. **rel_err** is $\|G_{\text{net}} - G_{\text{ode}}\|_F / \|G_{\text{net}}\|_F$.

index	epoch	rel_err	net loss	ODE loss
0	0	0.024126	259.1575	259.1575
12	48	0.258642	14.0089	116.0627
25	100	0.241244	0.7865	58.7949
38	152	0.228635	0.0000	32.5296
50	199	0.225354	0.0000	20.2713

Final relative Frobenius error, $N = 30$, $L = 2$, 5 seeds: mean $\pm$ std = 0.2156 ± 0.0377 .

C.6 $N = 30$, $L = 3$

Table 34: Seed 0: equivalence trajectory for $N = 30$, $L = 3$. **rel_err** is $\|G_{\text{net}} - G_{\text{ode}}\|_F / \|G_{\text{net}}\|_F$.

index	epoch	rel_err	net loss	ODE loss
0	0	0.017693	160.6104	160.6104
12	48	0.131446	14.9266	77.9091
25	100	0.216280	4.2152	44.9625
38	152	0.220143	4.0000	29.8939
50	199	0.218784	4.0000	20.6063

Table 35: Seed 1: equivalence trajectory for $N = 30$, $L = 3$. **rel_err** is $\|G_{\text{net}} - G_{\text{ode}}\|_F / \|G_{\text{net}}\|_F$.

index	epoch	rel_err	net loss	ODE loss
0	0	0.018608	174.4275	174.4275
12	48	0.096400	11.3126	86.0868
25	100	0.144113	2.8106	43.6845
38	152	0.150158	2.0000	24.9824
50	199	0.148644	2.0000	15.9909

Table 36: Seed 2: equivalence trajectory for $N = 30$, $L = 3$. **rel_err** is $\|G_{\text{net}} - G_{\text{ode}}\|_F / \|G_{\text{net}}\|_F$.

index	epoch	rel_err	net loss	ODE loss
0	0	0.014676	164.4582	164.4582
12	48	0.161947	12.3523	85.9750
25	100	0.252987	0.0797	51.8453
38	152	0.255928	0.0000	32.5765
50	199	0.254324	0.0000	19.7801

Table 37: Seed 3: equivalence trajectory for $N = 30$, $L = 3$. **rel_err** is $\|G_{\text{net}} - G_{\text{ode}}\|_F / \|G_{\text{net}}\|_F$.

index	epoch	rel_err	net loss	ODE loss
0	0	0.017828	167.4935	167.4935
12	48	0.132251	7.4126	85.0525
25	100	0.174941	0.0000	41.5284
38	152	0.171941	0.0000	23.5201
50	199	0.170278	0.0000	13.4652

Table 38: Seed 4: equivalence trajectory for $N = 30$, $L = 3$. **rel_err** is $\|G_{\text{net}} - G_{\text{ode}}\|_F / \|G_{\text{net}}\|_F$.

index	epoch	rel_err	net loss	ODE loss
0	0	0.015692	161.8157	161.8157
12	48	0.127256	12.8296	79.6979
25	100	0.195308	1.7468	44.5627
38	152	0.218119	0.0000	25.5578
50	199	0.217233	0.0000	17.0729

Final relative Frobenius error, $N = 30$, $L = 3$, 5 seeds: mean $\pm$ std = 0.2019 ± 0.0377 .

D Per-seed preconditioner effects

Per-seed log-log slopes for the spectral preconditioner experiments. For each seed and each variant (vanilla / preconditioned) we fit $\log |\tilde{G}_{ii}(T) - \tilde{G}_{ii}(0)|$ against $\log \lambda_i$ at the final recorded epoch.

D.1 Synthetic depth-1, $N = 50$ (`preconditioner_N50.json`)

Table 39: Per-seed slopes, synthetic $N = 50$, depth 1.

seed	variant	slope	R^2	$\lambda_{\min}$	$\lambda_{\max}$	modes
0	vanilla	1.9216	0.8939	3.531e-01	1.907e+01	50
0	precond	0.7585	0.7915	3.531e-01	1.907e+01	50
1	vanilla	1.8951	0.9189	3.583e-01	1.922e+01	50
1	precond	0.8295	0.8158	3.583e-01	1.922e+01	50
2	vanilla	1.9106	0.7913	3.582e-01	1.935e+01	50
2	precond	0.5606	0.4198	3.582e-01	1.935e+01	50
3	vanilla	2.1023	0.8889	3.621e-01	1.901e+01	50
3	precond	0.7936	0.6784	3.621e-01	1.901e+01	50
4	vanilla	2.1512	0.7815	3.416e-01	1.906e+01	50
4	precond	0.7352	0.7249	3.416e-01	1.906e+01	50
vanilla mean $\pm$ std		1.9962 ± 0.1081				
precond mean $\pm$ std		0.7355 ± 0.0931				

D.2 CIFAR-10 head-only (`cifar10.json`)

Table 40: Per-seed slopes, frozen-feature CIFAR-10 with depth-2 head.

seed	variant	slope	R^2	$\lambda_{\min}$	$\lambda_{\max}$	modes
0	vanilla	2.0656	0.9642	6.295e-01	2.834e+02	400
0	precond	1.5838	0.9702	6.295e-01	2.834e+02	400
1	vanilla	2.0635	0.9655	7.107e-01	2.829e+02	400
1	precond	1.6009	0.9653	7.107e-01	2.829e+02	400
2	vanilla	2.0374	0.9702	3.004e-01	2.830e+02	400
2	precond	1.5611	0.9655	3.004e-01	2.830e+02	400
3	vanilla	2.0583	0.9650	5.645e-01	2.831e+02	400
3	precond	1.5890	0.9657	5.645e-01	2.831e+02	400
4	vanilla	2.0646	0.9674	3.487e-01	2.831e+02	400
4	precond	1.6081	0.9667	3.487e-01	2.831e+02	400
vanilla mean $\pm$ std		2.0579 ± 0.0106				
precond mean $\pm$ std		1.5886 ± 0.0162				

D.3 CIFAR-10 end-to-end CNN (cifar10_end2end.json)

Table 41: Per-seed slopes, end-to-end CNN trained on CIFAR-10 from scratch.

seed	variant	slope	R^2	$\lambda_{\min}$	$\lambda_{\max}$	modes
0	vanilla	1.1206	0.6261	3.402e+00	2.584e+04	100
0	precond	1.0252	0.6956	3.402e+00	2.584e+04	100
1	vanilla	1.2510	0.7137	2.861e+00	6.960e+04	100
1	precond	1.0677	0.6602	2.861e+00	6.960e+04	100
2	vanilla	1.0523	0.6385	2.877e+00	4.613e+04	100
2	precond	0.7595	0.5409	2.877e+00	4.613e+04	100
3	vanilla	1.1785	0.7716	3.261e+00	5.692e+04	100
3	precond	0.9580	0.6725	3.261e+00	5.692e+04	100
4	vanilla	1.1365	0.6573	3.366e+00	8.366e+04	100
4	precond	1.0372	0.6733	3.366e+00	8.366e+04	100
vanilla mean $\pm$ std		1.1478 $\pm$ 0.0657				
precond mean $\pm$ std		0.9695 $\pm$ 0.1110				

D.4 Triplet satisfaction (synthetic $N = 50$)

Table 42: Per-seed overall triplet satisfaction and per-bin satisfaction for synthetic $N = 50$.

seed	overall vanilla	overall precond	bin0 (vanilla / precond)	bin1 (vanilla / precond)
0	0.8300	0.8100	0.8305 / 0.8017	0.8269 / 0.8654
1	0.8325	0.8025	0.8275 / 0.7953	0.8621 / 0.8448
2	0.8300	0.8100	0.8373 / 0.8136	0.7903 / 0.7903
3	0.8200	0.8550	0.8099 / 0.8421	0.8727 / 0.9273
4	0.8375	0.7800	0.8418 / 0.7712	0.8043 / 0.8478

E Recall@K: full per-seed curves

Held-out CIFAR-10 retrieval metrics for both triplet-difficulty regimes. For each of $K \in \{1, 5, 10, 20\}$ we report all five seeds’ vanilla and preconditioned recall, the per-seed difference $\Delta = \text{precond} - \text{vanilla}$, and a paired t -statistic / two-sided p -value over the five paired differences.

E.1 Easy triplets

Table 43: Easy triplets, recall@1 per seed.

seed	vanilla	precond	Δ
0	0.7450	0.7300	-0.0150
1	0.7300	0.7200	-0.0100
2	0.6800	0.6550	-0.0250
3	0.6900	0.7000	0.0100
4	0.7300	0.6850	-0.0450
mean $\pm$ std	0.7150 ± 0.0253	0.6980 ± 0.0266	-0.0170 ± 0.0181
paired t / p	$t = -1.883$, two-sided $p = 0.1328$		

Table 44: Easy triplets, recall@5 per seed.

seed	vanilla	precond	Δ
0	0.9150	0.9150	0.0000
1	0.9300	0.9300	0.0000
2	0.8750	0.9000	0.0250
3	0.9050	0.9050	0.0000
4	0.9250	0.9300	0.0050
mean $\pm$ std	0.9100 ± 0.0195	0.9160 ± 0.0124	0.0060 ± 0.0097
paired t / p	$t = 1.238$, two-sided $p = 0.2835$		

Table 45: Easy triplets, recall@10 per seed.

seed	vanilla	precond	Δ
0	0.9650	0.9700	0.0050
1	0.9600	0.9550	-0.0050
2	0.9250	0.9350	0.0100
3	0.9400	0.9450	0.0050
4	0.9700	0.9650	-0.0050
mean $\pm$ std	0.9520 ± 0.0169	0.9540 ± 0.0128	0.0020 ± 0.0060
paired t / p	$t = 0.667$, two-sided $p = 0.5415$		

Table 46: Easy triplets, recall@20 per seed.

seed	vanilla	precond	Δ
0	0.9850	0.9850	0.0000
1	0.9900	0.9950	0.0050
2	0.9600	0.9650	0.0050
3	0.9650	0.9700	0.0050
4	0.9850	0.9900	0.0050
mean $\pm$ std	0.9770 ± 0.0121	0.9810 ± 0.0116	0.0040 ± 0.0020
paired t / p	$t = 4.000$, two-sided $p = 0.0161$		

E.2 Hard triplets

Table 47: Hard triplets, recall@1 per seed.

seed	vanilla	precond	Δ
0	0.7400	0.7300	-0.0100
1	0.7450	0.7150	-0.0300
2	0.7250	0.7200	-0.0050
3	0.7100	0.7450	0.0350
4	0.7450	0.7300	-0.0150
mean $\pm$ std	0.7330 ± 0.0136	0.7280 ± 0.0103	-0.0050 ± 0.0217
paired t / p	$t = -0.461$, two-sided $p = 0.6686$		

Table 48: Hard triplets, recall@5 per seed.

seed	vanilla	precond	Δ
0	0.8650	0.8900	0.0250
1	0.9000	0.9100	0.0100
2	0.8600	0.8700	0.0100
3	0.8550	0.8800	0.0250
4	0.8950	0.9350	0.0400
mean $\pm$ std	0.8750 ± 0.0187	0.8970 ± 0.0232	0.0220 ± 0.0112
paired t / p	$t = 3.920$, two-sided $p = 0.0173$		

Table 49: Hard triplets, recall@10 per seed.

seed	vanilla	precond	Δ
0	0.9250	0.9650	0.0400
1	0.9350	0.9500	0.0150
2	0.9000	0.9250	0.0250
3	0.9050	0.9500	0.0450
4	0.9300	0.9650	0.0350
mean $\pm$ std	0.9190 ± 0.0139	0.9510 ± 0.0146	0.0320 ± 0.0108
paired t / p	$t = 5.942$, two-sided $p = 0.0040$		

Table 50: Hard triplets, recall@20 per seed.

seed	vanilla	precond	Δ
0	0.9600	0.9850	0.0250
1	0.9700	0.9850	0.0150
2	0.9400	0.9650	0.0250
3	0.9500	0.9550	0.0050
4	0.9550	0.9950	0.0400
mean $\pm$ std	0.9550 ± 0.0100	0.9770 ± 0.0147	0.0220 ± 0.0117
paired t / p	$t = 3.773$, two-sided $p = 0.0196$		

F Phase-transition full trajectory

Effective rank and triplet loss as a function of the nuclear-norm regularization strength λ , sweeping λ across 25 log-spaced points. We tabulate every λ value and report the per-seed rank and loss, plus the cross-seed mean $\pm$ std.

Table 51: Phase-transition sweep; per-seed rank and loss.

λ	seed 0 (rank/loss)	seed 1	seed 2	seed 3	seed 4	rank mean $\pm$ std	loss mean $\pm$ std
0.1000	10/23.53	10/11.03	10/12.83	10/8.66	10/15.32	10.00 ± 0.00	14.275 ± 5.116
0.1334	10/24.79	10/12.57	10/14.19	10/10.11	10/16.62	10.00 ± 0.00	15.656 ± 5.037
0.1778	10/26.30	10/14.52	10/15.86	10/11.84	10/18.38	10.00 ± 0.00	17.380 ± 4.933
0.2371	10/28.03	10/17.04	10/17.96	10/13.84	10/20.58	10.00 ± 0.00	19.487 ± 4.781
0.3162	10/29.98	10/20.01	10/20.36	10/16.15	10/23.15	10.00 ± 0.00	21.929 ± 4.601
0.4217	10/32.16	10/23.23	10/22.98	10/18.75	10/26.12	10.00 ± 0.00	24.649 ± 4.432
0.5623	10/34.45	10/26.63	10/25.91	10/21.57	10/29.60	10.00 ± 0.00	27.632 ± 4.266
0.7499	10/36.74	10/30.15	10/28.91	10/24.72	10/33.03	10.00 ± 0.00	30.710 ± 4.032
1.0000	10/38.71	10/33.73	10/31.96	10/28.51	10/36.12	10.00 ± 0.00	33.806 ± 3.489
1.3335	10/40.68	9/37.37	10/35.09	10/32.31	10/39.24	9.80 ± 0.40	36.938 ± 2.976
1.7783	10/42.86	9/40.29	10/38.24	10/35.81	10/42.20	9.80 ± 0.40	39.879 ± 2.596
2.3714	10/45.17	9/43.26	8/41.58	9/39.58	9/45.47	9.00 ± 0.63	43.014 ± 2.218

λ	seed 0 (rank/loss)	seed 1	seed 2	seed 3	seed 4	rank mean $\pm$ std	loss mean $\pm$ std
3.1623	8/47.52	8/46.09	8/45.33	8/43.13	8/48.44	8.00 ± 0.00	46.104 ± 1.839
4.2170	6/49.36	6/48.67	7/48.26	8/47.77	7/49.36	6.80 ± 0.75	48.684 ± 0.620
5.6234	4/50.08	5/50.82	5/49.17	5/50.19	5/49.92	4.80 ± 0.40	50.037 ± 0.531
7.4989	3/50.47	5/50.54	4/50.48	3/50.67	5/50.40	4.00 ± 0.89	50.513 ± 0.092
10.0000	4/50.08	4/50.10	5/50.07	3/50.08	6/50.06	4.40 ± 1.02	50.077 ± 0.016
13.3352	4/50.01	3/50.02	5/50.01	4/50.00	5/50.01	4.20 ± 0.75	50.009 ± 0.005
17.7828	4/50.00	2/50.00	3/50.00	3/50.00	2/50.00	2.80 ± 0.75	50.001 ± 0.001
23.7137	2/50.00	1/50.00	2/50.00	3/50.00	2/50.00	2.00 ± 0.63	50.000 ± 0.000
31.6228	5/50.00	1/50.00	5/50.00	10/50.00	2/50.00	4.60 ± 3.14	50.000 ± 0.000
42.1697	10/50.00	5/50.00	9/50.00	10/50.00	8/50.00	8.40 ± 1.85	50.000 ± 0.000
56.2341	10/50.00	9/50.00	9/50.00	10/50.00	10/50.00	9.60 ± 0.49	50.000 ± 0.000
74.9894	10/50.00	10/50.00	9/50.00	10/50.00	10/50.00	9.80 ± 0.40	50.000 ± 0.000
100.0000	10/50.00	10/50.00	9/50.00	10/50.00	10/50.00	9.80 ± 0.40	50.000 ± 0.000

G Rank-vs-width: full sweep

For each network width M in the sweep, the learned effective rank, final triplet loss, and triplet-satisfaction rate. $N = 10$ for this experiment, so $M \geq N$ regimes test the capacity-saturated case.

Table 52: Rank, loss, satisfaction by width and seed.

M	seed 0	seed 1	seed 2	seed 3	seed 4	rank mean $\pm$ std	loss mean $\pm$ std
5	5/43.85	4/45.34	4/34.54	5/32.89	4/29.29	4.40 ± 0.49	37.182 ± 6.304
8	5/38.05	5/37.83	5/26.45	6/31.61	7/22.19	5.60 ± 0.80	31.228 ± 6.240
10	8/35.27	7/34.00	6/27.02	7/28.23	6/17.33	6.80 ± 0.75	28.370 ± 6.374
20	8/18.55	8/27.61	9/22.62	10/21.40	8/17.50	8.60 ± 0.80	21.537 ± 3.556
50	10/13.31	10/21.27	10/15.17	10/12.79	10/10.76	10.00 ± 0.00	14.659 ± 3.591
100	10/10.76	10/12.46	10/10.84	10/3.49	10/3.17	10.00 ± 0.00	8.144 ± 3.977
200	10/3.99	10/10.43	10/3.99	10/2.43	10/2.95	10.00 ± 0.00	4.760 ± 2.899

Table 53: Triplet satisfaction by width and seed.

M	seed 0	seed 1	seed 2	seed 3	seed 4	sat mean $\pm$ std
5	0.000	0.000	0.080	0.280	0.200	0.112 ± 0.111
8	0.040	0.060	0.320	0.140	0.420	0.196 ± 0.149
10	0.140	0.160	0.120	0.180	0.400	0.200 ± 0.102
20	0.500	0.260	0.280	0.300	0.340	0.336 ± 0.086
50	0.500	0.420	0.500	0.540	0.600	0.512 ± 0.059
100	0.660	0.600	0.560	0.740	0.780	0.668 ± 0.083
200	0.800	0.700	0.740	0.860	0.920	0.804 ± 0.079

H Rank-vs-spectral trajectory

Per-epoch rank and Frobenius-norm spectral displacement on the same trajectory. The synthetic experiment uses $N = 50$, depth 1, width 1000. `spectral_diff` columns are $\|\tilde{G}(t) - \tilde{G}(0)\|_F$ for the indicated variant.

Seed 0

Table 54: Seed 0: rank trajectory and spectral displacement, $N = 50$ depth-1.

epoch	rank V	rank P	spec_diff V	spec_diff P	loss V	loss P
0	34	34	0.6172	0.0865	1525.490	1525.490
200	35	34	82.5049	11.8430	766.222	1042.834
400	35	34	121.5688	13.5623	505.931	802.163
600	35	34	144.9039	11.0663	374.378	645.949
800	35	34	160.1366	10.4735	290.312	544.838
1000	35	34	171.4824	17.0497	230.349	468.652
1200	35	34	180.0846	27.6956	183.610	409.107
1400	35	34	186.6353	40.7013	147.832	360.203
1600	35	34	191.5418	54.7752	120.546	320.041
1800	34	34	195.4396	68.7049	98.353	287.829
1999	34	34	198.3005	82.7277	80.741	261.172

Seed 1

Table 55: Seed 1: rank trajectory and spectral displacement, $N = 50$ depth-1.

epoch	rank V	rank P	spec_diff V	spec_diff P	loss V	loss P
0	34	34	0.5655	0.0563	1601.171	1601.171
200	34	33	73.0666	7.2196	790.609	1070.974
400	34	33	110.5828	11.3019	517.979	826.971
600	34	33	133.3961	17.6226	366.871	677.349
800	33	32	149.2934	27.9538	271.307	570.327
1000	33	32	160.2255	42.3889	203.534	487.096
1200	33	32	167.2141	58.4366	154.976	420.643
1400	33	32	172.0935	76.1353	118.798	366.148
1600	32	32	176.0684	94.6835	91.526	323.095
1800	32	32	179.0111	113.5556	70.299	286.876
1999	32	32	180.8209	132.1042	53.259	256.191

Seed 2

Table 56: Seed 2: rank trajectory and spectral displacement, $N = 50$ depth-1.

epoch	rank V	rank P	spec_diff V	spec_diff P	loss V	loss P
0	33	33	0.5233	0.0792	1346.290	1346.290
200	33	33	67.9436	7.8463	701.309	928.787
400	34	34	103.5779	9.9446	458.184	721.717
600	34	34	126.7586	11.1524	318.878	580.250
800	34	33	141.0613	13.7513	238.916	475.457
1000	34	33	150.1412	20.1901	187.155	393.118
1200	34	33	156.3735	29.8339	148.399	331.761
1400	34	33	160.7937	41.3218	119.992	283.308
1600	34	33	164.1770	53.4519	98.973	242.781
1800	34	33	167.1063	66.4719	82.531	210.367
1999	34	33	169.4178	79.7799	68.691	183.047

Seed 3

Table 57: Seed 3: rank trajectory and spectral displacement, $N = 50$ depth-1.

epoch	rank V	rank P	spec_diff V	spec_diff P	loss V	loss P
0	32	32	0.4729	0.0435	1375.993	1375.993
200	33	33	63.0315	4.4401	704.348	921.898
400	34	33	97.2550	6.9618	463.811	716.447
600	34	33	117.7547	11.6179	330.192	580.987
800	34	33	131.0592	19.8268	251.243	480.930
1000	34	33	140.3624	30.3525	195.627	405.190
1200	34	33	147.2359	42.8376	156.165	346.855
1400	34	32	152.1893	57.3204	123.874	299.984
1600	34	32	156.2254	73.5763	99.484	258.744
1800	33	32	159.4999	89.8526	79.858	225.378
1999	33	32	161.9305	105.5890	64.403	197.469

Seed 4

Table 58: Seed 4: rank trajectory and spectral displacement, $N = 50$ depth-1.

epoch	rank V	rank P	spec_diff V	spec_diff P	loss V	loss P
0	34	34	0.5754	0.0401	1556.637	1556.637
200	35	33	74.8543	6.3150	756.306	1094.554
400	35	34	110.0039	12.6049	483.438	836.149
600	34	34	131.4728	20.3611	338.145	672.589
800	34	34	145.6640	29.8972	250.000	557.935
1000	34	33	155.2717	41.2984	188.619	474.704
1200	34	33	161.6604	54.3247	142.388	410.328
1400	33	33	165.8882	68.9000	107.062	357.808
1600	33	33	168.9079	84.4378	81.105	314.431
1800	33	33	171.1840	101.1100	63.042	276.315
1999	33	33	172.8506	118.0078	49.708	243.981

Cross-seed summary at final epoch

Table 59: Final-epoch cross-seed summary.

metric	vanilla	precond
effective rank	33.20 ± 0.75	32.80 ± 0.75
spectral displacement	176.664 ± 12.406	103.642 ± 20.135
final loss	63.360 ± 11.136	228.372 ± 31.946

I NTK eigenvalue spectra

For each configuration where the empirical NTK was diagonalized, we list the leading and trailing eigenvalues at the standard seed (seed = 0). Eigenvalues are sorted in decreasing order. For $N \leq 20$ we list all eigenvalues; otherwise we list the top 10 and bottom 5 with the count of intermediate eigenvalues.

$N = 30, L = 1$ (synthetic) (**n = 30**)

Table 60: Eigenvalue spectrum for $N = 30, L = 1$ (synthetic).

index (sorted desc)	λ
0	1.1980e+01
1	4.3442e+00
2	4.2283e+00
3	3.6581e+00
4	3.3864e+00
5	3.0575e+00
6	2.6957e+00
7	2.5667e+00
8	2.3622e+00
9	2.2249e+00
... (15 intermediate eigenvalues) ...	
25	6.2271e-01
26	5.6583e-01
27	5.2397e-01
28	4.9450e-01
29	4.5498e-01
$\lambda_{\max}/\lambda_{\min,+}$	2.633e+01
effective rank $\text{tr}(K)^2/\ K\ _F^2$	13.740

$N = 100$, $L = 1$ (synthetic) ($n = 100$)

Table 61: Eigenvalue spectrum for $N = 100$, $L = 1$ (synthetic).

index (sorted desc)	λ
0	3.6393e+01
1	1.0198e+01
2	9.3329e+00
3	8.9916e+00
4	8.3349e+00
5	8.0303e+00
6	7.7562e+00
7	6.6264e+00
8	6.0757e+00
9	5.8788e+00
... (85 intermediate eigenvalues) ...	
95	2.8899e-01
96	2.7473e-01
97	2.7097e-01
98	2.5754e-01
99	2.4680e-01
$\lambda_{\max}/\lambda_{\min,+}$	1.475e+02
effective rank $\text{tr}(K)^2/\ K\ _F^2$	18.951

$N = 1000$, $L = 1$ (synthetic) (**n = 1000**)

Table 62: Eigenvalue spectrum for $N = 1000$, $L = 1$ (synthetic).

index (sorted desc)	λ
0	3.4437e+02
1	6.2154e+01
2	6.0907e+01
3	5.8557e+01
4	5.7720e+01
5	5.6053e+01
6	5.4974e+01
7	5.3693e+01
8	5.3358e+01
9	5.1219e+01
... (985 intermediate eigenvalues) ...	
995	1.4980e-01
996	1.4955e-01
997	1.4920e-01
998	1.4868e-01
999	1.4843e-01
$\lambda_{\max}/\lambda_{\min,+}$	2.320e+03
effective rank $\text{tr}(K)^2/\ K\ _F^2$	23.248

$N = 30, L = 2$ (synthetic) (**n = 30**)

Table 63: Eigenvalue spectrum for $N = 30, L = 2$ (synthetic).

index (sorted desc)	λ
0	2.3739e+01
1	4.9203e+00
2	4.8453e+00
3	4.2884e+00
4	4.0348e+00
5	3.6797e+00
6	3.3360e+00
7	3.1696e+00
8	2.9586e+00
9	2.8461e+00
... (15 intermediate eigenvalues) ...	
25	1.2497e+00
26	1.1824e+00
27	1.1142e+00
28	1.0869e+00
29	1.0146e+00
$\lambda_{\max}/\lambda_{\min,+}$	2.340e+01
effective rank $\text{tr}(K)^2/\ K\ _F^2$	10.748

$N = 100$, $L = 2$ (synthetic) ($n = 100$)

Table 64: Eigenvalue spectrum for $N = 100$, $L = 2$ (synthetic).

index (sorted desc)	λ
0	7.4433e+01
1	1.0728e+01
2	9.8539e+00
3	9.5401e+00
4	8.8719e+00
5	8.5650e+00
6	8.3030e+00
7	7.1879e+00
8	6.6814e+00
9	6.4551e+00
... (85 intermediate eigenvalues) ...	
95	7.9900e-01
96	7.8087e-01
97	7.7612e-01
98	7.5764e-01
99	7.5552e-01
$\lambda_{\max}/\lambda_{\min,+}$	9.852e+01
effective rank $\text{tr}(K)^2/\ K\ _F^2$	13.710

$N = 1000$, $L = 2$ (synthetic) (**n = 1000**)

Table 65: Eigenvalue spectrum for $N = 1000$, $L = 2$ (synthetic).

index (sorted desc)	λ
0	7.1954e+02
1	6.1885e+01
2	6.0634e+01
3	5.8378e+01
4	5.7521e+01
5	5.5866e+01
6	5.4786e+01
7	5.3579e+01
8	5.3212e+01
9	5.1076e+01
... (985 intermediate eigenvalues) ...	
995	4.9050e-01
996	4.8897e-01
997	4.8826e-01
998	4.8650e-01
999	4.8254e-01
$\lambda_{\max}/\lambda_{\min,+}$	1.491e+03
effective rank $\text{tr}(K)^2/\ K\ _F^2$	15.709

Fashion-MNIST $N = 200$ depth-2 (n = 200)

Table 66: Eigenvalue spectrum for Fashion-MNIST $N = 200$ depth-2.

index (sorted desc)	λ
0	1.5834e+02
1	6.1743e+01
2	4.2369e+01
3	2.7472e+01
4	1.7932e+01
5	1.0932e+01
6	1.0015e+01
7	9.8283e+00
8	7.8122e+00
9	7.0282e+00
... (185 intermediate eigenvalues) ...	
195	3.2311e-01
196	3.0566e-01
197	3.0367e-01
198	3.0067e-01
199	2.8169e-01
$\lambda_{\max}/\lambda_{\min,+}$	5.621e+02
effective rank $\text{tr}(K)^2/\ K\ _F^2$	10.994

CIFAR-10 head-only $N = 400$ depth-2 (n = 400)

Table 67: Eigenvalue spectrum for CIFAR-10 head-only $N = 400$ depth-2.

index (sorted desc)	λ
0	2.8342e+02
1	5.2672e+01
2	2.3462e+01
3	2.1119e+01
4	1.9599e+01
5	1.5114e+01
6	1.4599e+01
7	1.2356e+01
8	1.1843e+01
9	1.0792e+01
... (385 intermediate eigenvalues) ...	
395	7.7658e-01
396	7.5549e-01
397	7.3938e-01
398	7.0104e-01
399	6.2946e-01
$\lambda_{\max}/\lambda_{\min,+}$	4.503e+02
effective rank $\text{tr}(K)^2/\ K\ _F^2$	16.441

CIFAR-10 end-to-end CNN $N = 100$ ($n = 100$)

Table 68: Eigenvalue spectrum for CIFAR-10 end-to-end CNN $N = 100$.

index (sorted desc)	λ
0	2.5842e+04
1	3.4146e+03
2	9.2564e+02
3	5.2422e+02
4	4.6629e+02
5	3.2167e+02
6	2.4081e+02
7	2.1379e+02
8	1.8525e+02
9	1.5380e+02
... (85 intermediate eigenvalues) ...	
95	3.9942e+00
96	3.9313e+00
97	3.7567e+00
98	3.5426e+00
99	3.4023e+00
$\lambda_{\max}/\lambda_{\min,+}$	7.596e+03
effective rank $\text{tr}(K)^2/\ K\ _F^2$	1.698

synthetic $N = 50$ depth-1 ($n = 50$)

Table 69: Eigenvalue spectrum for synthetic $N = 50$ depth-1.

index (sorted desc)	λ
0	1.9071e+01
1	6.6175e+00
2	5.8636e+00
3	5.5983e+00
4	4.5533e+00
5	4.3064e+00
6	4.0600e+00
7	3.7496e+00
8	3.3934e+00
9	3.0404e+00
... (35 intermediate eigenvalues) ...	
45	4.6447e-01
46	4.3790e-01
47	4.1888e-01
48	3.9411e-01
49	3.5309e-01
$\lambda_{\max}/\lambda_{\min,+}$	5.401e+01
effective rank $\text{tr}(K)^2/\ K\ _F^2$	16.107

J Per-mode displacement tables

For the smaller configurations where listing every mode is feasible, we report λ_i alongside the per-seed final displacement $|\tilde{G}_{ii}(T) - \tilde{G}_{ii}(0)|$. Larger configurations are summarized by listing the top-10 and bottom-10 modes only, since the same information is captured by the slope/ R^2 of Appendix B.

J.1 $N = 30, L = 1$ (full)

Table 70: Per-mode displacement, $N = 30, L = 1$. Columns are seed-0 eigenvalues then $|\tilde{G}_{ii}(T) - \tilde{G}_{ii}(0)|$ for each seed.

i	$\lambda_i(s0)$	disp s0	disp s1	disp s2	disp s3	disp s4
0	1.1980e+01	6.833e+01	6.829e+01	6.588e+01	9.839e+01	1.069e+02
1	4.3442e+00	2.145e+01	2.960e+01	1.738e+01	4.073e+01	3.301e+01
2	4.2283e+00	2.997e+01	2.293e+01	1.775e+01	2.798e+01	3.354e+01
3	3.6581e+00	2.264e+01	1.755e+01	1.883e+01	2.232e+01	2.773e+01
4	3.3864e+00	1.767e+01	1.608e+01	1.954e+01	1.129e+01	1.580e+01
5	3.0575e+00	7.191e+00	1.036e+01	6.786e+00	8.658e+00	1.288e+01
6	2.6957e+00	7.204e+00	1.058e+01	7.740e+00	1.307e+01	1.202e+01
7	2.5667e+00	5.943e+00	1.553e+01	5.971e+00	5.055e+00	9.344e+00

i	$\lambda_i(s0)$	disp s0	disp s1	disp s2	disp s3	disp s4
8	2.3622e+00	1.228e+00	5.696e+00	3.924e+00	1.155e+01	1.029e+01
9	2.2249e+00	3.760e+00	3.770e+00	6.787e-01	1.080e+01	1.051e+01
10	2.0267e+00	1.419e+00	1.687e+00	3.336e+00	3.905e+00	2.039e+00
11	1.7828e+00	1.523e+00	4.592e-01	3.748e+00	2.948e+00	5.385e+00
12	1.7242e+00	2.748e+00	2.175e+00	4.211e+00	1.844e+00	2.806e+00
13	1.5574e+00	1.390e+00	1.374e+00	8.149e-02	2.867e+00	1.380e+00
14	1.3383e+00	1.764e+00	3.414e+00	1.297e+00	1.679e-01	6.417e-01
15	1.1242e+00	4.688e-01	3.181e+00	1.171e+00	9.982e-01	1.145e+00
16	9.8741e-01	2.414e-01	2.116e+00	3.108e-01	4.444e-01	2.397e+00
17	9.7102e-01	9.937e-02	1.378e-02	1.215e+00	6.873e-01	9.160e-01
18	8.7564e-01	1.691e+00	7.458e-01	5.242e-01	1.419e+00	1.214e-01
19	8.2643e-01	1.422e+00	3.562e-01	3.053e-01	1.757e+00	3.463e-01
20	7.7609e-01	3.578e-01	6.442e-01	7.022e-01	1.451e+00	7.153e-02
21	7.4691e-01	4.100e-01	6.756e-01	3.234e-01	9.462e-01	3.804e-01
22	7.4049e-01	8.465e-02	4.039e-01	5.396e-01	1.848e-01	2.722e-02
23	6.9564e-01	2.870e-01	1.803e+00	3.602e-01	1.080e+00	1.275e-01
24	6.6119e-01	1.150e-01	3.049e-01	4.492e-02	5.720e-01	7.192e-01
25	6.2271e-01	2.040e-01	6.059e-02	1.477e-01	6.414e-01	1.861e+00
26	5.6583e-01	6.991e-01	2.907e-02	9.828e-02	3.663e-01	4.241e-02
27	5.2397e-01	2.670e-01	1.337e-01	6.214e-01	4.221e-02	1.173e+00
28	4.9450e-01	8.426e-02	3.656e-01	3.757e-02	2.177e-01	4.955e-02
29	4.5498e-01	7.374e-01	2.867e-01	1.018e+00	1.157e-01	6.860e-01

J.2 $N = 30, L = 2$ (full)

Table 71: Per-mode displacement, $N = 30, L = 2$. Columns are seed-0 eigenvalues then $|\tilde{G}_{ii}(T) - \tilde{G}_{ii}(0)|$ for each seed.

i	$\lambda_i(s0)$	disp s0	disp s1	disp s2	disp s3	disp s4
0	2.3739e+01	1.407e+01	3.408e+01	2.118e+01	4.274e+01	4.040e+01
1	4.9203e+00	5.019e+00	9.219e+00	4.260e+00	1.383e+01	8.647e+00
2	4.8453e+00	8.987e+00	1.047e+01	3.161e+00	8.430e+00	1.090e+01
3	4.2884e+00	6.964e+00	3.599e+00	5.755e+00	7.545e+00	8.652e+00
4	4.0348e+00	5.433e+00	4.319e+00	6.471e+00	3.684e+00	2.944e+00
5	3.6797e+00	1.133e+00	2.638e+00	1.963e+00	2.710e+00	1.524e+00
6	3.3360e+00	6.532e-01	4.442e+00	1.126e+00	2.715e+00	4.435e+00
7	3.1696e+00	9.965e-01	6.677e+00	1.119e+00	6.112e-01	1.028e+00
8	2.9586e+00	5.848e-02	1.700e+00	1.402e+00	3.263e+00	3.721e+00
9	2.8461e+00	7.304e-01	1.873e+00	8.158e-01	3.793e+00	2.624e+00
10	2.6526e+00	1.394e+00	1.260e+00	4.005e-01	7.487e-01	2.820e-01
11	2.4306e+00	1.244e+00	1.399e+00	1.055e+00	8.225e-01	9.910e-01
12	2.3455e+00	4.831e-02	1.829e+00	9.454e-01	1.119e-01	4.914e-01
13	2.2054e+00	2.050e-01	2.843e-01	8.004e-03	2.364e+00	2.688e-01

i	$\lambda_i(s_0)$	disp s0	disp s1	disp s2	disp s3	disp s4
14	2.0075e+00	1.955e-01	1.287e+00	1.633e+00	9.449e-01	8.363e-01
15	1.7545e+00	8.610e-01	1.503e+00	2.560e-01	4.139e-01	1.160e-01
16	1.6449e+00	8.295e-01	6.025e-01	6.530e-03	4.347e-01	2.823e-01
17	1.6346e+00	5.596e-01	3.222e-02	4.107e-01	2.012e-01	1.443e+00
18	1.5265e+00	1.406e+00	6.658e-01	3.455e-01	5.246e-01	1.173e+00
19	1.4690e+00	1.277e+00	5.393e-01	2.333e-01	1.182e+00	5.572e-01
20	1.4447e+00	5.076e-01	1.842e+00	5.795e-01	3.816e-01	8.558e-01
21	1.4045e+00	4.464e-01	6.610e-01	1.132e+00	1.478e-02	1.478e-01
22	1.3866e+00	4.148e-01	3.669e-01	1.105e+00	2.161e-01	1.090e+00
23	1.3270e+00	7.169e-02	1.330e+00	1.912e-01	9.610e-01	8.227e-01
24	1.3000e+00	7.847e-01	5.184e-01	7.260e-01	7.831e-02	8.676e-02
25	1.2497e+00	4.563e-01	2.423e-02	3.567e-01	8.050e-01	8.209e-01
26	1.1824e+00	5.639e-01	5.425e-01	3.499e-01	3.883e-01	4.558e-01
27	1.1142e+00	5.677e-01	5.191e-02	7.944e-01	4.254e-01	1.217e+00
28	1.0869e+00	2.888e-02	7.735e-01	1.697e-01	4.933e-01	4.763e-01
29	1.0146e+00	7.679e-01	5.973e-01	6.431e-01	3.007e-01	7.757e-01

J.3 $N = 100, L = 1$ (top/bottom 10)

Table 72: $N = 100, L = 1$: top-10 and bottom-10 eigenmodes by seed-0 ordering.

rank	idx	$\lambda(s_0)$	disp s0	disp s1	disp s2	disp s3	disp s4
top 0	0	3.6393e+01	8.162e+02	8.574e+02	8.183e+02	7.452e+02	7.517e+02
top 1	1	1.0198e+01	1.631e+02	1.711e+02	1.607e+02	1.589e+02	1.771e+02
top 2	2	9.3329e+00	1.463e+02	1.511e+02	1.485e+02	1.496e+02	1.484e+02
top 3	3	8.9916e+00	1.345e+02	1.463e+02	1.230e+02	1.061e+02	1.276e+02
top 4	4	8.3349e+00	1.363e+02	1.246e+02	1.464e+02	1.197e+02	1.387e+02
top 5	5	8.0303e+00	1.302e+02	1.027e+02	1.046e+02	8.061e+01	9.808e+01
top 6	6	7.7562e+00	1.128e+02	1.148e+02	8.781e+01	9.021e+01	9.234e+01
top 7	7	6.6264e+00	8.622e+01	9.145e+01	8.665e+01	5.834e+01	8.437e+01
top 8	8	6.0757e+00	9.664e+01	7.610e+01	5.970e+01	6.107e+01	6.891e+01
top 9	9	5.8788e+00	5.130e+01	7.927e+01	7.857e+01	6.242e+01	5.956e+01
bot 0	90	3.2642e-01	6.843e-01	8.650e-01	6.193e-01	8.079e-01	9.688e-01
bot 1	91	3.1782e-01	7.269e-01	9.723e-01	7.534e-01	6.602e-01	7.327e-01
bot 2	92	3.1486e-01	5.756e-01	7.024e-01	5.882e-01	6.528e-01	8.661e-01
bot 3	93	3.0286e-01	9.979e-01	6.883e-01	5.447e-01	5.174e-01	7.720e-01
bot 4	94	2.9312e-01	6.798e-01	8.044e-01	4.777e-01	5.692e-01	7.037e-01
bot 5	95	2.8899e-01	9.875e-01	7.016e-01	7.205e-01	6.568e-01	6.362e-01
bot 6	96	2.7473e-01	5.870e-01	7.826e-01	6.324e-01	6.665e-01	6.488e-01
bot 7	97	2.7097e-01	5.767e-01	7.266e-01	6.773e-01	4.356e-01	6.032e-01
bot 8	98	2.5754e-01	5.868e-01	6.290e-01	4.856e-01	6.068e-01	5.284e-01
bot 9	99	2.4680e-01	3.678e-01	7.011e-01	1.669e-01	5.699e-01	5.339e-01

J.4 $N = 100$, $L = 2$ (top/bottom 10)

Table 73: $N = 100$, $L = 2$: top-10 and bottom-10 eigenmodes by seed-0 ordering.

rank	idx	$\lambda(s_0)$	disp s0	disp s1	disp s2	disp s3	disp s4
top 0	0	7.4433e+01	6.825e+02	7.059e+02	6.438e+02	6.916e+02	6.464e+02
top 1	1	1.0728e+01	5.502e+01	5.472e+01	5.620e+01	6.151e+01	6.445e+01
top 2	2	9.8539e+00	5.481e+01	5.549e+01	4.499e+01	5.276e+01	4.881e+01
top 3	3	9.5401e+00	4.459e+01	5.149e+01	3.724e+01	3.586e+01	4.423e+01
top 4	4	8.8719e+00	4.369e+01	4.549e+01	4.483e+01	4.104e+01	4.837e+01
top 5	5	8.5650e+00	4.284e+01	3.669e+01	3.735e+01	3.026e+01	3.171e+01
top 6	6	8.3030e+00	3.912e+01	3.870e+01	3.354e+01	3.195e+01	3.196e+01
top 7	7	7.1879e+00	3.255e+01	3.606e+01	2.986e+01	2.554e+01	3.031e+01
top 8	8	6.6814e+00	3.524e+01	3.002e+01	2.637e+01	2.636e+01	2.774e+01
top 9	9	6.4551e+00	2.412e+01	3.194e+01	3.051e+01	2.489e+01	2.429e+01
bot 0	90	8.6157e-01	5.454e-01	3.712e-01	4.464e-01	5.614e-01	3.079e-01
bot 1	91	8.4728e-01	5.706e-01	4.695e-01	4.151e-01	1.769e-01	3.334e-01
bot 2	92	8.3622e-01	2.292e-01	3.720e-01	3.556e-01	3.346e-01	3.648e-01
bot 3	93	8.2014e-01	4.101e-01	3.003e-01	1.249e-01	3.642e-01	3.584e-01
bot 4	94	8.1246e-01	3.670e-01	3.472e-01	3.549e-01	1.689e-01	3.179e-01
bot 5	95	7.9900e-01	3.830e-01	2.788e-01	3.974e-01	2.915e-01	2.614e-01
bot 6	96	7.8087e-01	4.976e-01	2.954e-01	3.458e-01	4.140e-01	4.092e-01
bot 7	97	7.7612e-01	4.542e-01	3.351e-01	2.841e-01	3.346e-01	3.589e-01
bot 8	98	7.5764e-01	3.863e-01	3.452e-01	2.550e-02	3.305e-01	2.184e-01
bot 9	99	7.5552e-01	2.352e-01	3.284e-01	2.383e-01	3.821e-01	2.556e-01

J.5 $N = 1000$, $L = 1$ (top/bottom 10)

Table 74: $N = 1000$, $L = 1$: top-10 and bottom-10 eigenmodes by seed-0 ordering.

rank	idx	$\lambda(s_0)$	disp s0	disp s1	disp s2	disp s3	disp s4
top 0	0	3.4437e+02	1.489e+04	1.489e+04	1.462e+04	1.460e+04	1.471e+04
top 1	1	6.2154e+01	1.458e+03	1.569e+03	1.436e+03	1.518e+03	1.378e+03
top 2	2	6.0907e+01	1.310e+03	1.419e+03	1.386e+03	1.357e+03	1.394e+03
top 3	3	5.8557e+01	1.361e+03	1.277e+03	1.263e+03	1.350e+03	1.366e+03
top 4	4	5.7720e+01	1.266e+03	1.378e+03	1.339e+03	1.325e+03	1.255e+03
top 5	5	5.6053e+01	1.253e+03	1.261e+03	1.308e+03	1.216e+03	1.304e+03
top 6	6	5.4974e+01	1.287e+03	1.371e+03	1.294e+03	1.251e+03	1.291e+03
top 7	7	5.3693e+01	1.222e+03	1.253e+03	1.227e+03	1.190e+03	1.218e+03
top 8	8	5.3358e+01	1.203e+03	1.179e+03	1.176e+03	1.138e+03	1.188e+03
top 9	9	5.1219e+01	1.132e+03	1.238e+03	1.183e+03	1.161e+03	1.128e+03
bot 0	990	1.5090e-01	5.668e-01	6.231e-01	5.316e-01	5.608e-01	5.479e-01
bot 1	991	1.5073e-01	5.468e-01	5.472e-01	5.490e-01	5.133e-01	5.530e-01
bot 2	992	1.5066e-01	5.826e-01	5.773e-01	5.985e-01	5.376e-01	5.822e-01
bot 3	993	1.5030e-01	5.633e-01	5.450e-01	5.455e-01	5.781e-01	5.535e-01
bot 4	994	1.4998e-01	5.668e-01	5.435e-01	5.501e-01	5.345e-01	5.602e-01
bot 5	995	1.4980e-01	6.034e-01	5.609e-01	5.838e-01	5.439e-01	5.322e-01
bot 6	996	1.4955e-01	5.681e-01	5.760e-01	5.423e-01	5.577e-01	5.712e-01
bot 7	997	1.4920e-01	5.620e-01	5.207e-01	5.409e-01	5.278e-01	5.485e-01
bot 8	998	1.4868e-01	5.595e-01	5.777e-01	5.329e-01	5.262e-01	5.672e-01
bot 9	999	1.4843e-01	5.321e-01	5.475e-01	5.327e-01	5.447e-01	5.553e-01

J.6 $N = 1000$, $L = 2$ (top/bottom 10)

Table 75: $N = 1000$, $L = 2$: top-10 and bottom-10 eigenmodes by seed-0 ordering.

rank	idx	$\lambda(s_0)$	disp s_0	disp s_1	disp s_2	disp s_3	disp s_4
top 0	0	7.1954e+02	7.063e+03	7.280e+03	7.370e+03	7.140e+03	6.864e+03
top 1	1	6.1885e+01	3.818e+02	4.398e+02	3.815e+02	4.063e+02	3.800e+02
top 2	2	6.0634e+01	3.698e+02	3.583e+02	3.836e+02	3.665e+02	3.814e+02
top 3	3	5.8378e+01	3.820e+02	3.444e+02	3.350e+02	3.491e+02	3.607e+02
top 4	4	5.7521e+01	3.500e+02	3.740e+02	3.747e+02	3.717e+02	3.360e+02
top 5	5	5.5866e+01	3.227e+02	3.428e+02	3.386e+02	2.950e+02	3.353e+02
top 6	6	5.4786e+01	3.586e+02	3.833e+02	3.760e+02	3.597e+02	3.419e+02
top 7	7	5.3579e+01	3.129e+02	3.270e+02	3.261e+02	3.469e+02	3.058e+02
top 8	8	5.3212e+01	3.261e+02	3.181e+02	3.224e+02	2.652e+02	3.019e+02
top 9	9	5.1076e+01	2.687e+02	3.396e+02	3.232e+02	3.462e+02	3.077e+02
bot 0	990	4.9580e-01	2.228e-01	2.850e-01	2.524e-01	1.843e-01	2.156e-01
bot 1	991	4.9493e-01	2.486e-01	2.289e-01	2.427e-01	2.237e-01	2.836e-01
bot 2	992	4.9414e-01	2.430e-01	2.384e-01	2.122e-01	2.637e-01	2.269e-01
bot 3	993	4.9356e-01	2.165e-01	2.339e-01	2.244e-01	2.402e-01	1.866e-01
bot 4	994	4.9201e-01	2.285e-01	2.268e-01	2.203e-01	1.828e-01	1.681e-01
bot 5	995	4.9050e-01	2.447e-01	1.841e-01	2.200e-01	2.454e-01	2.158e-01
bot 6	996	4.8897e-01	2.095e-01	2.212e-01	2.697e-01	2.298e-01	1.824e-01
bot 7	997	4.8826e-01	2.103e-01	2.043e-01	2.669e-01	2.174e-01	2.203e-01
bot 8	998	4.8650e-01	1.832e-01	2.772e-01	1.604e-01	2.403e-01	1.730e-01
bot 9	999	4.8254e-01	2.574e-01	2.351e-01	1.968e-01	2.100e-01	2.198e-01

K End-to-end CNN training details

Configuration (from JSON)

Table 76: End-to-end CIFAR-10 CNN training configuration.

key	value
N	100
n_classes	10
M_out	32
n_triplets	400
n_epochs	1200
lr	1e-06
margin	1.0
record_every	100
eps	0.01
data_root	./data

Per-epoch losses, all seeds

Table 77: End-to-end CIFAR-10 CNN per-epoch losses.

epoch	van s0	pre s0	van s1	pre s1	van s2	pre s2	van s3	pre s3	van s4	pre s4
0	891.205	891.205	2106.966	2106.966	802.178	802.178	1732.498	1732.498	1623.506	1623.506
100	324.500	873.434	306.762	2094.602	334.080	798.085	340.527	1716.405	317.976	1600.106
200	295.110	860.898	277.711	2087.295	309.199	793.970	303.921	1710.595	284.642	1586.846
300	271.124	848.851	257.864	2084.196	281.993	789.642	279.976	1706.481	259.959	1574.193
400	246.671	838.095	240.646	2087.463	257.591	785.507	260.632	1709.538	235.779	1563.187
500	222.499	828.348	222.869	2096.772	232.122	782.471	242.355	1713.987	217.109	1542.751
600	197.380	819.983	207.645	2112.104	210.900	779.608	225.998	1729.226	202.598	1515.682
700	178.045	814.327	194.076	2130.343	190.588	778.207	210.488	1741.576	190.454	1497.013
800	162.781	810.154	181.207	2152.861	173.055	776.404	196.101	1751.811	179.151	1481.223
900	150.368	807.700	169.408	2178.211	156.788	775.567	183.269	1762.599	167.906	1469.474
1000	138.779	807.179	159.568	2205.305	141.383	770.480	171.060	1773.667	156.563	1461.913
1100	129.158	807.027	150.419	2236.365	125.302	762.438	159.535	1787.418	145.380	1461.713
1199	120.513	808.231	142.510	2269.509	111.302	758.978	149.131	1798.283	134.476	1467.972

Per-seed final slopes

Table 78: End-to-end CIFAR-10 CNN: per-seed slopes and R^2 .

seed	vanilla slope	vanilla R^2	van modes	precond slope	precond R^2	pre modes
0	1.1206	0.6261	100	1.0252	0.6956	100
1	1.2510	0.7137	100	1.0677	0.6602	100
2	1.0523	0.6385	100	0.7595	0.5409	100
3	1.1785	0.7716	100	0.9580	0.6725	100
4	1.1365	0.6573	100	1.0372	0.6733	100
mean $\pm$ std	1.1478 $\pm$ 0.0657			0.9695 $\pm$ 0.1110		